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(54) ELECTRIC VEHICLE CHARGING STATION

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B60L 55/00	(2019.01)
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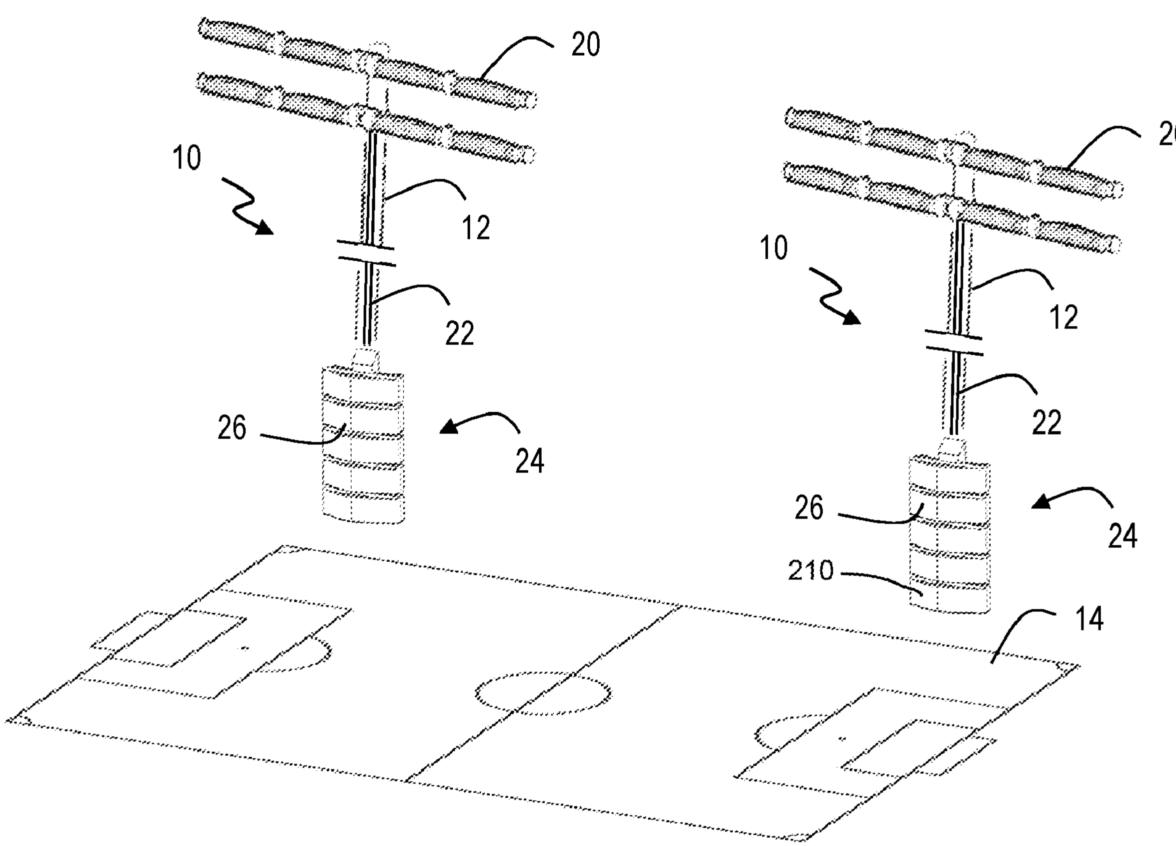
F21S 8/08	(2006.01)
F21S 9/02	(2006.01)
H02J 3/32	(2006.01)

(52) **U.S. Cl.**

CPC ***F21V 33/006*** (2013.01); ***B60L 53/18*** (2019.02); ***B60L 53/53*** (2019.02); ***B60L 53/67*** (2019.02); ***B60L 53/68*** (2019.02); ***B60L 55/00*** (2019.02); ***F21S 2/005*** (2013.01); ***F21S 8/086*** (2013.01); ***F21S 9/02*** (2013.01); ***H02J 3/322*** (2020.01)

(57) **ABSTRACT**

An electrical vehicle charging system having a lighting system having a controller stack connected to a lighting module, an auxiliary lighting module by wiring, an alternating current power supply, a switching circuit, a local power storage device, and a wireless communication gateway. The electrical vehicle charging system further having a plurality of electrical vehicle chargers each having an electrical vehicle charging cable configured for connection to an electrical vehicle; a connection to an existing power grid; an internet of things module configured for two-way, wireless communication with the wireless gateway; a feedback system; and a control system connected to and configured for communication with the internet of things module, the existing power grid, and the electrical vehicle charging cable.



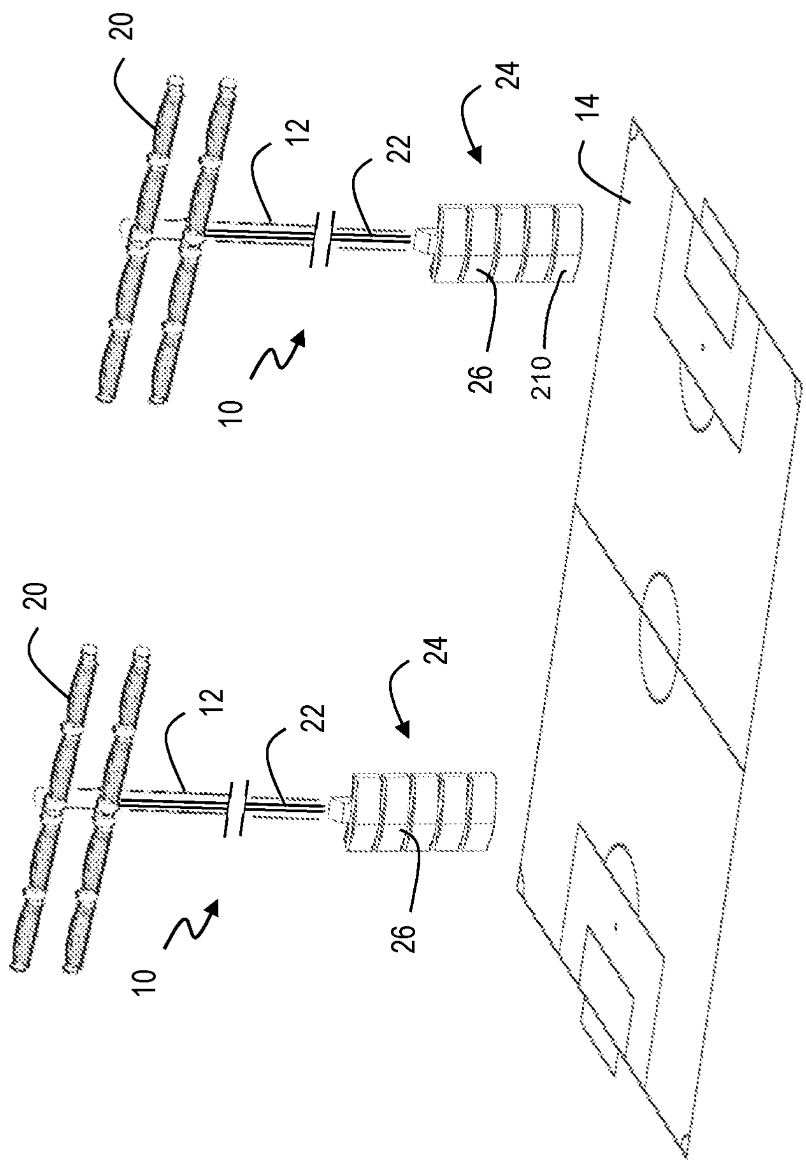


FIG. 1

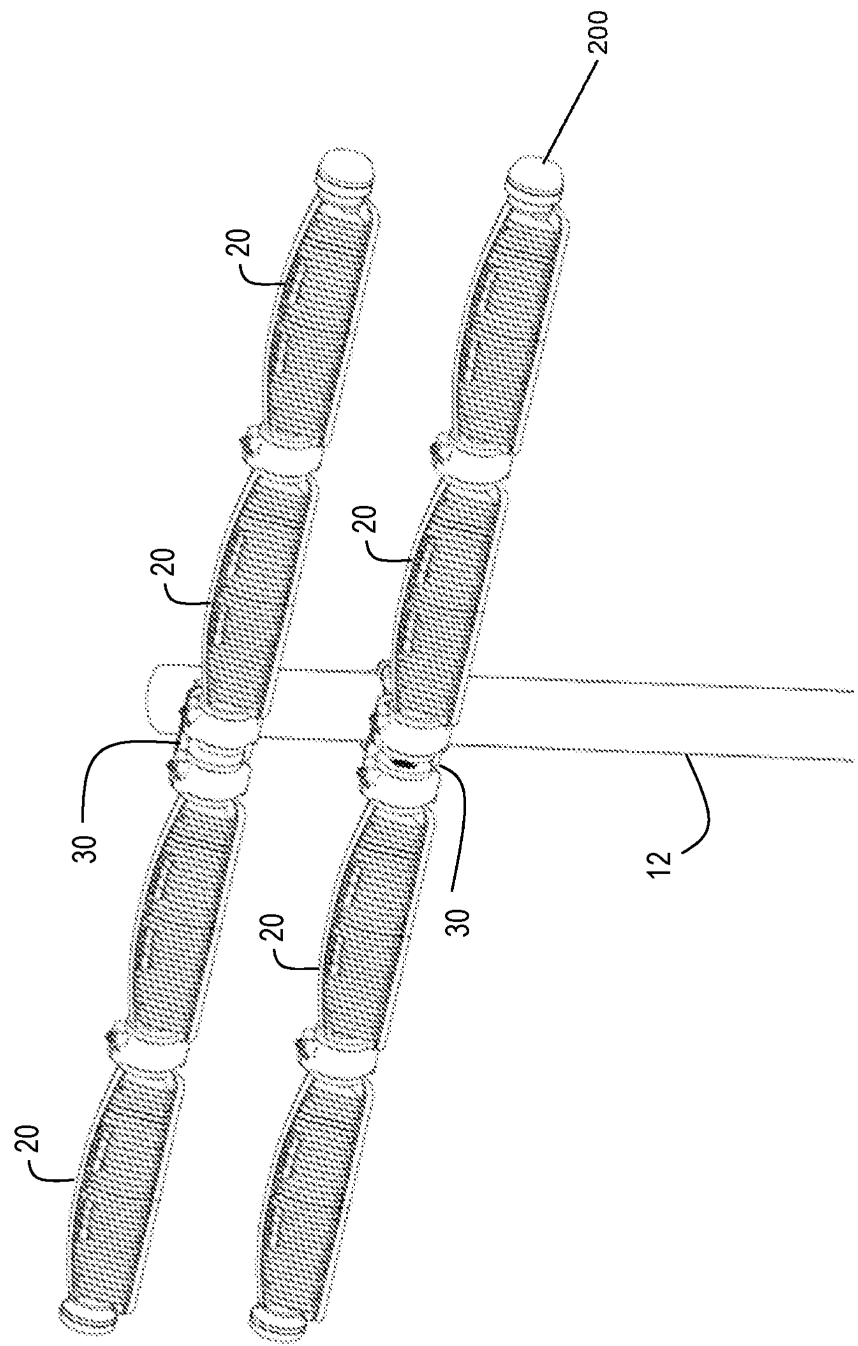


FIG. 2

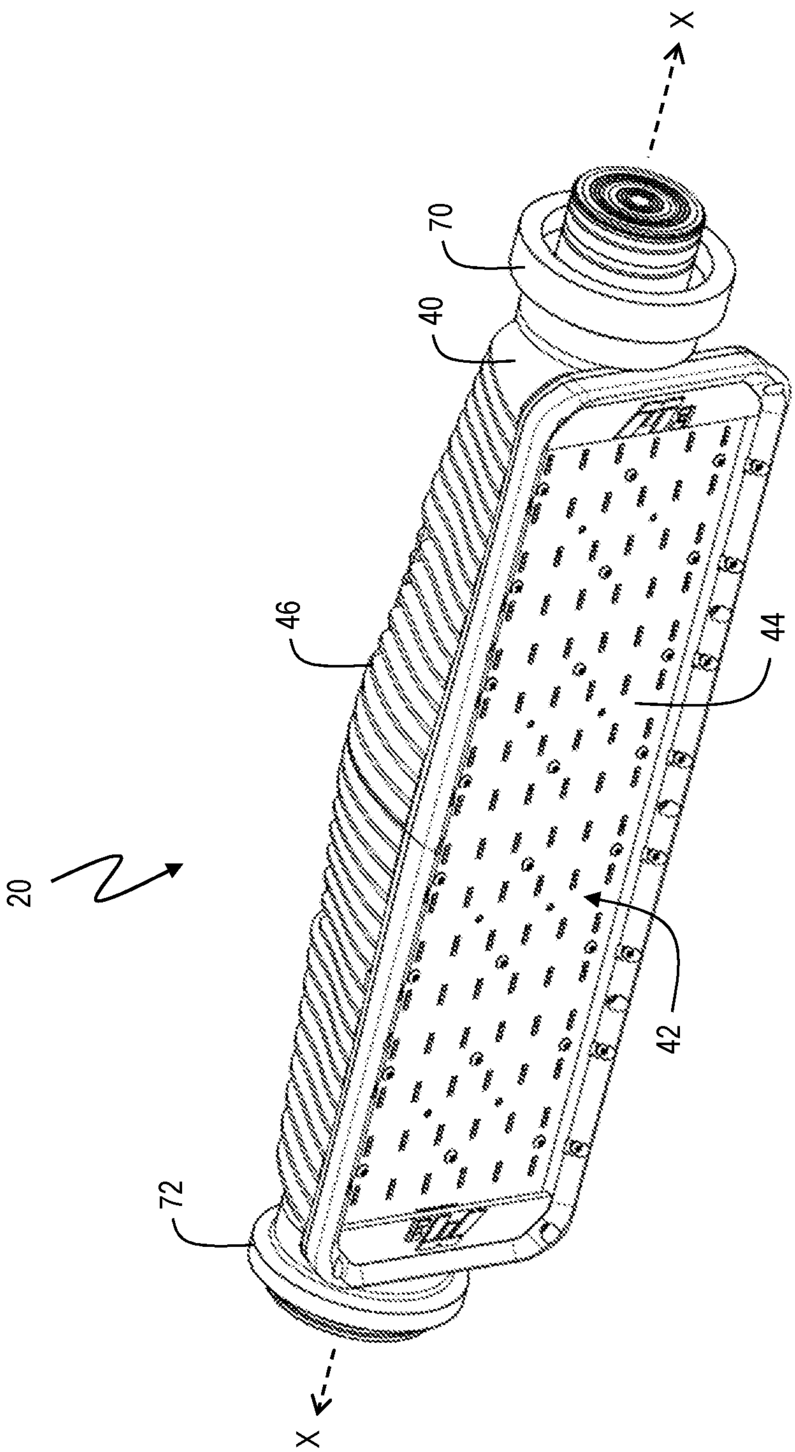


FIG. 3

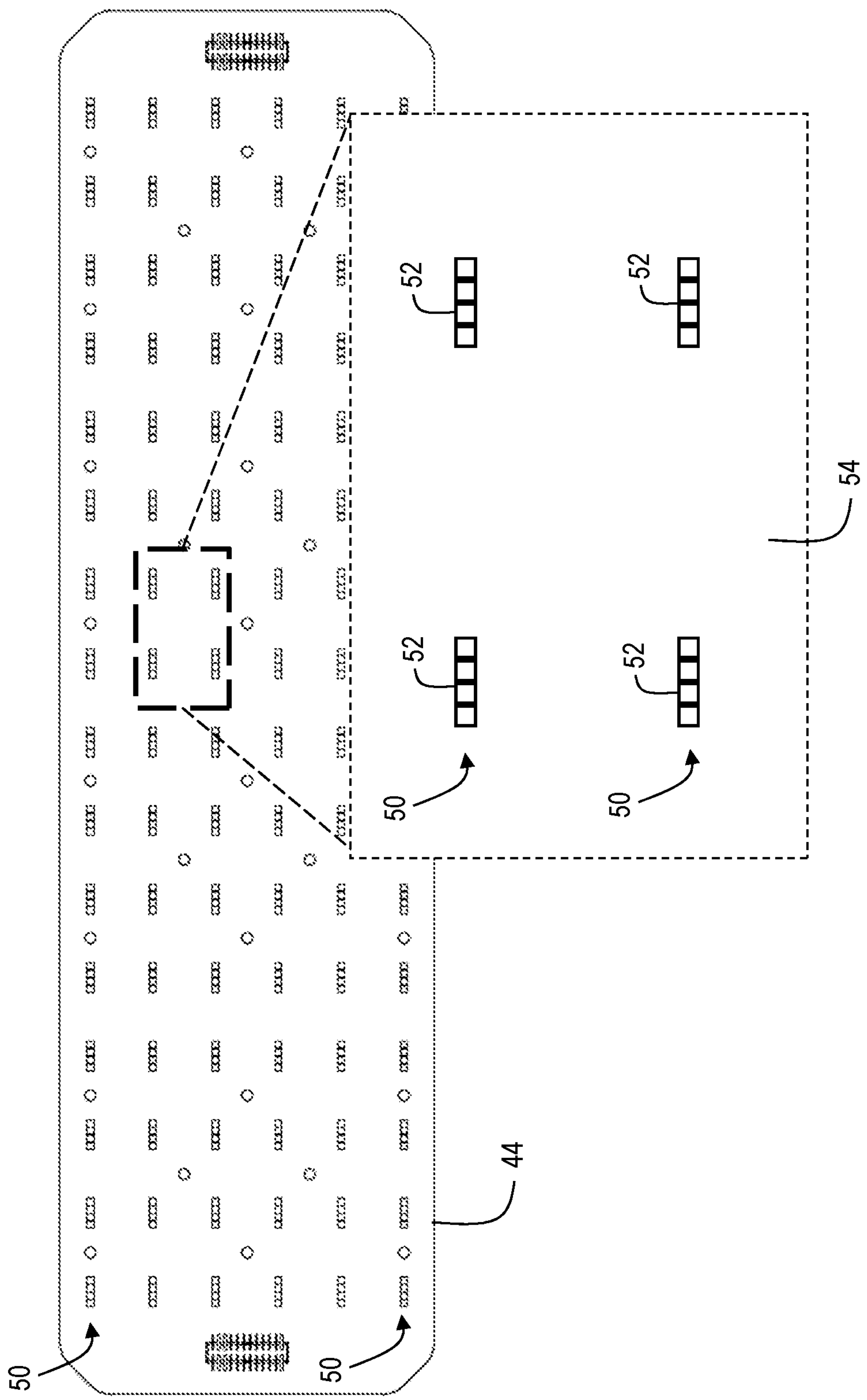


FIG. 4

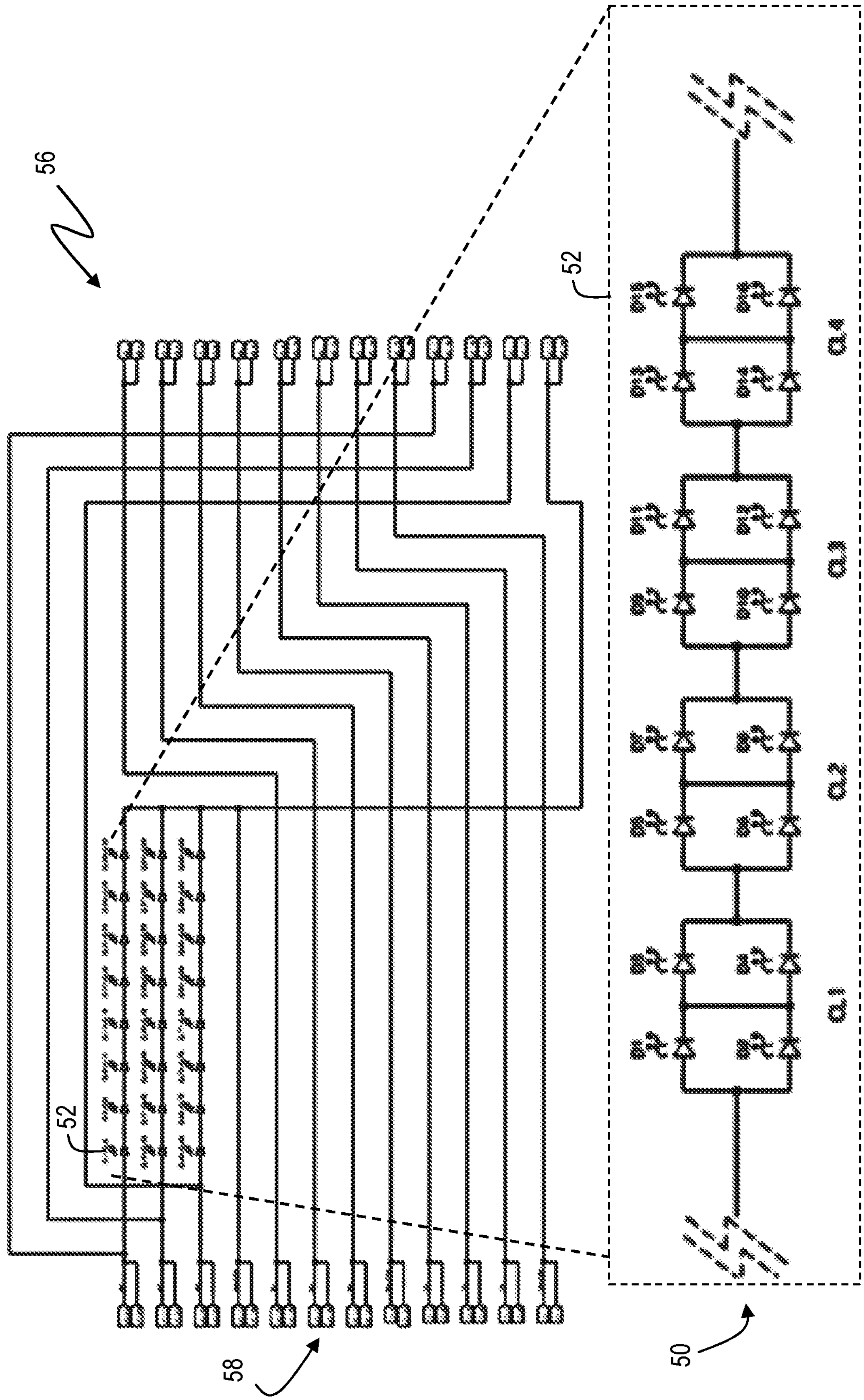


FIG. 5

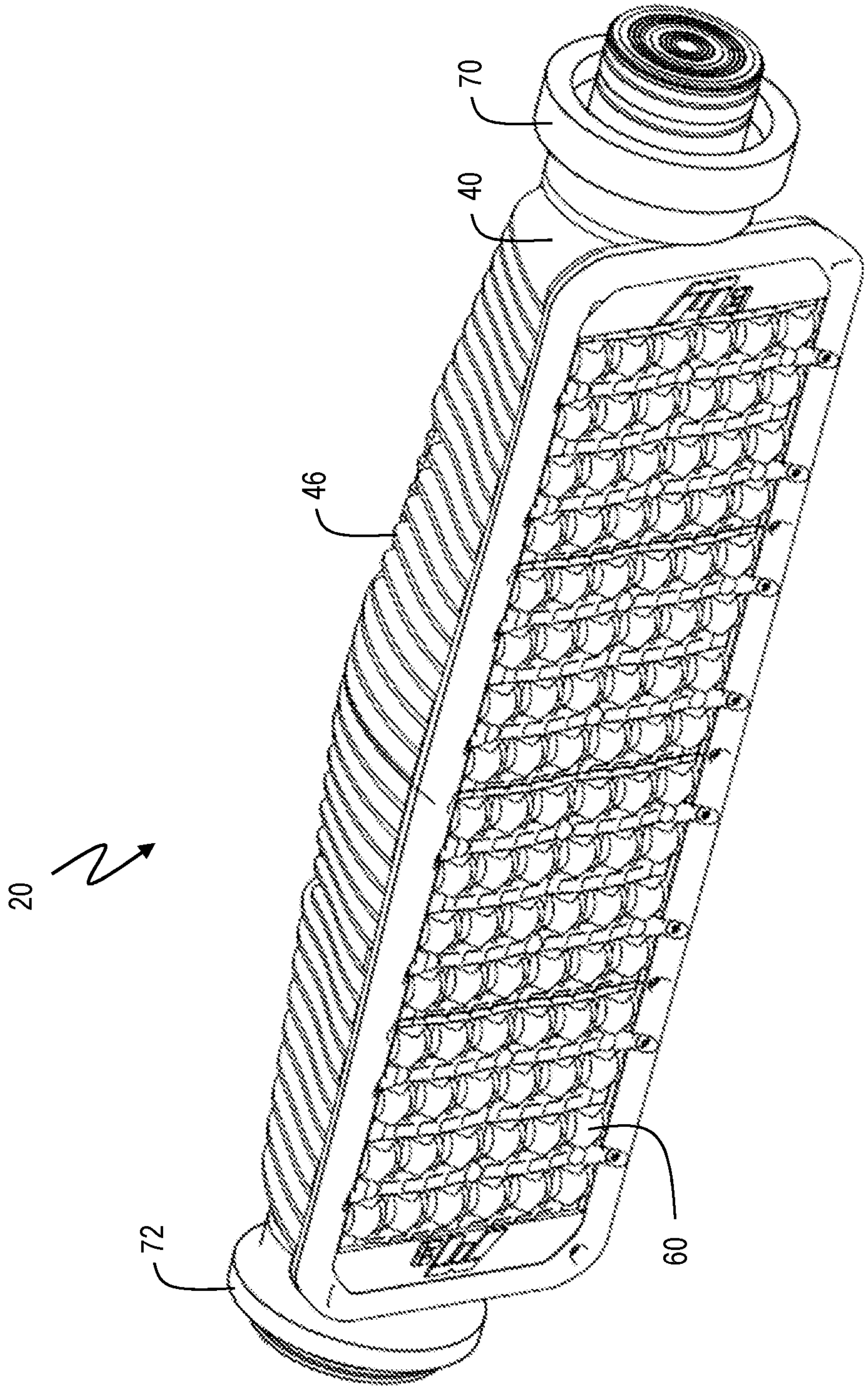


FIG. 6

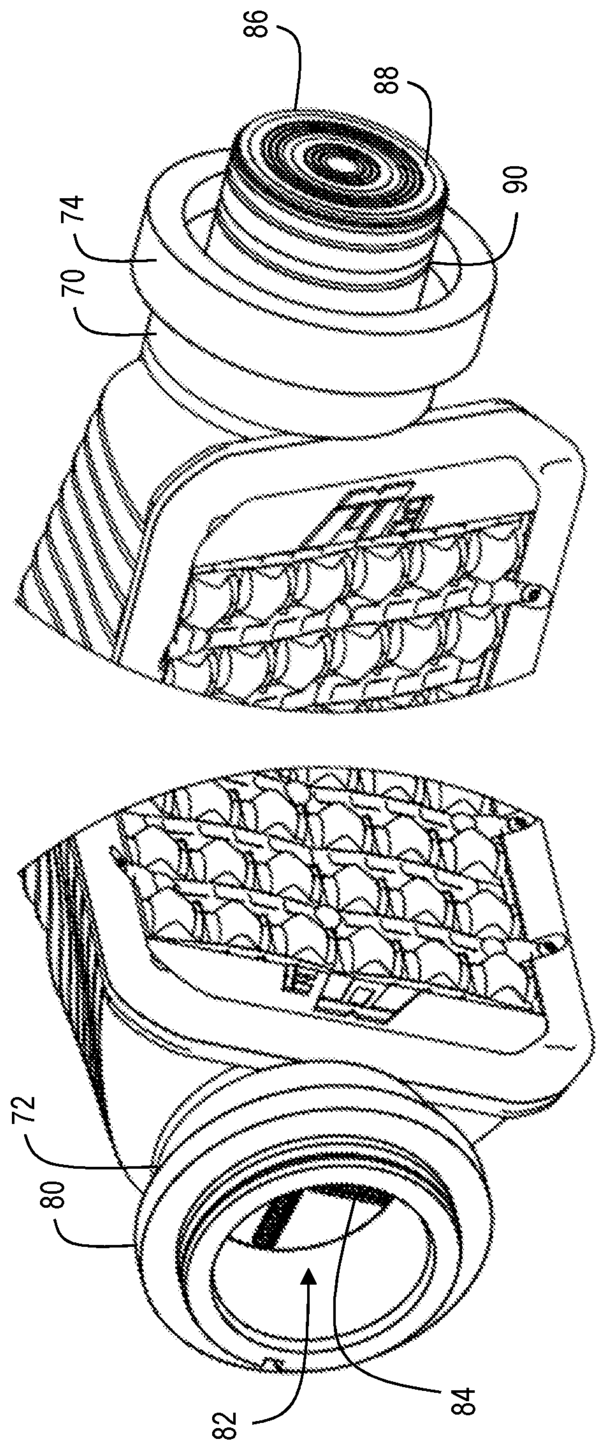


FIG. 7

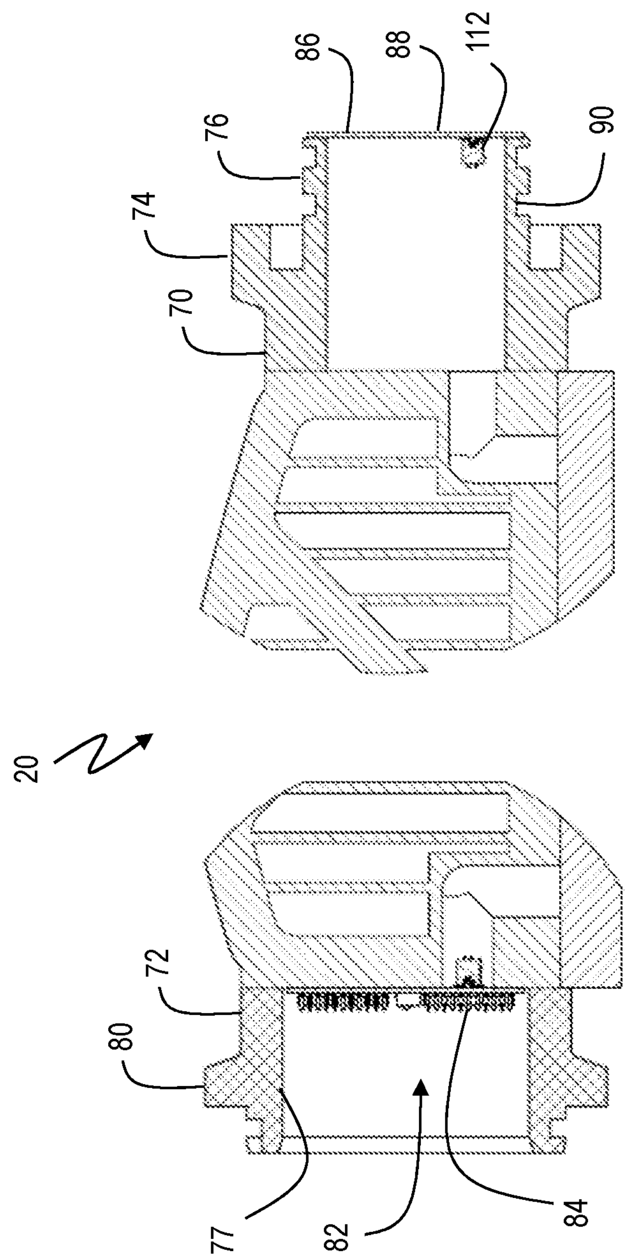


FIG. 8

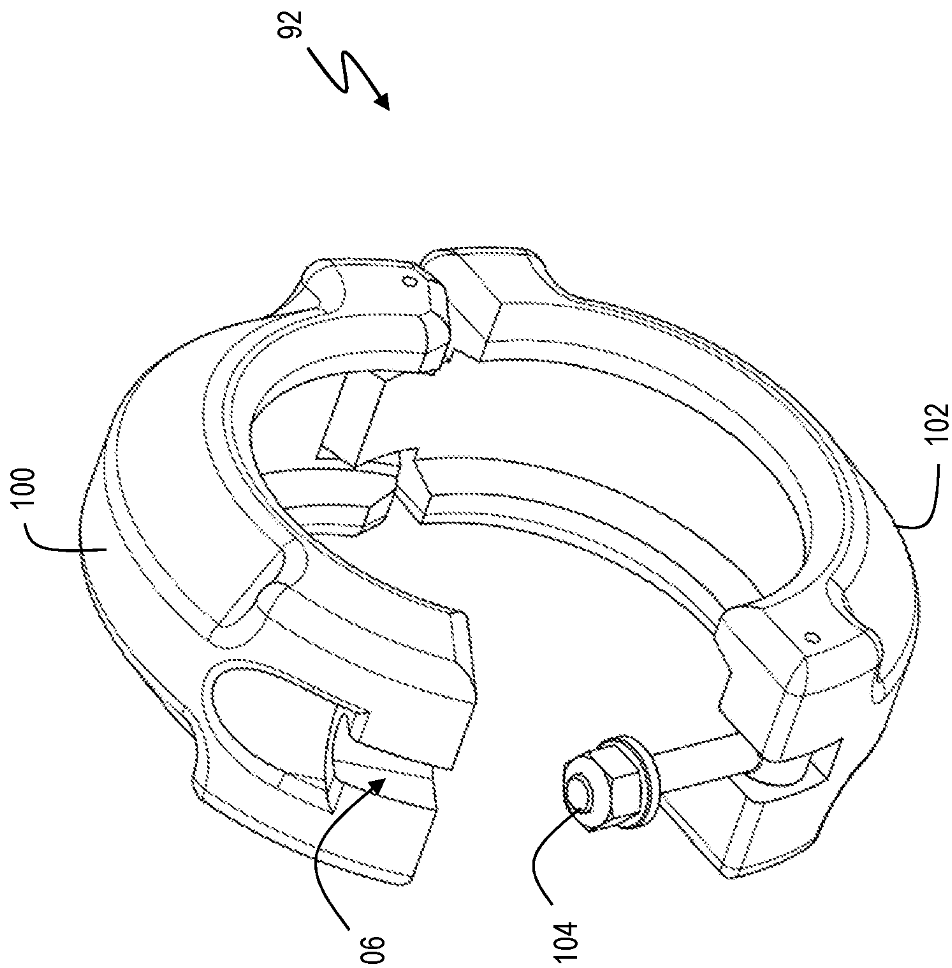


FIG. 9

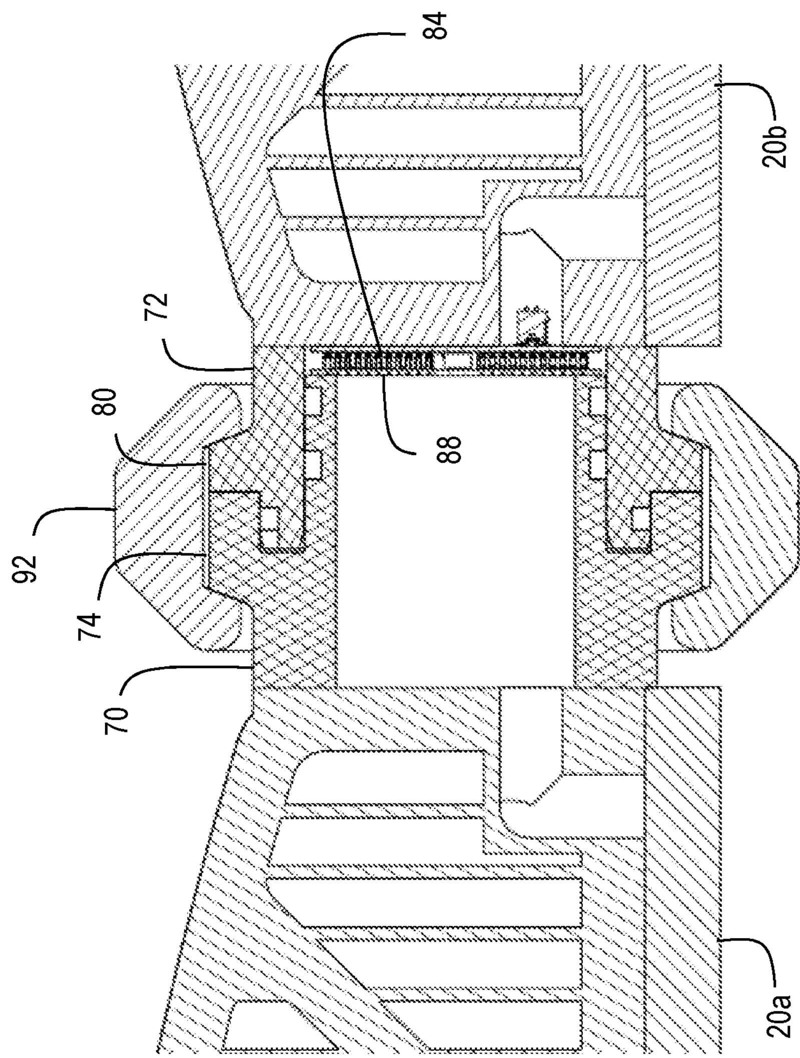


FIG. 10

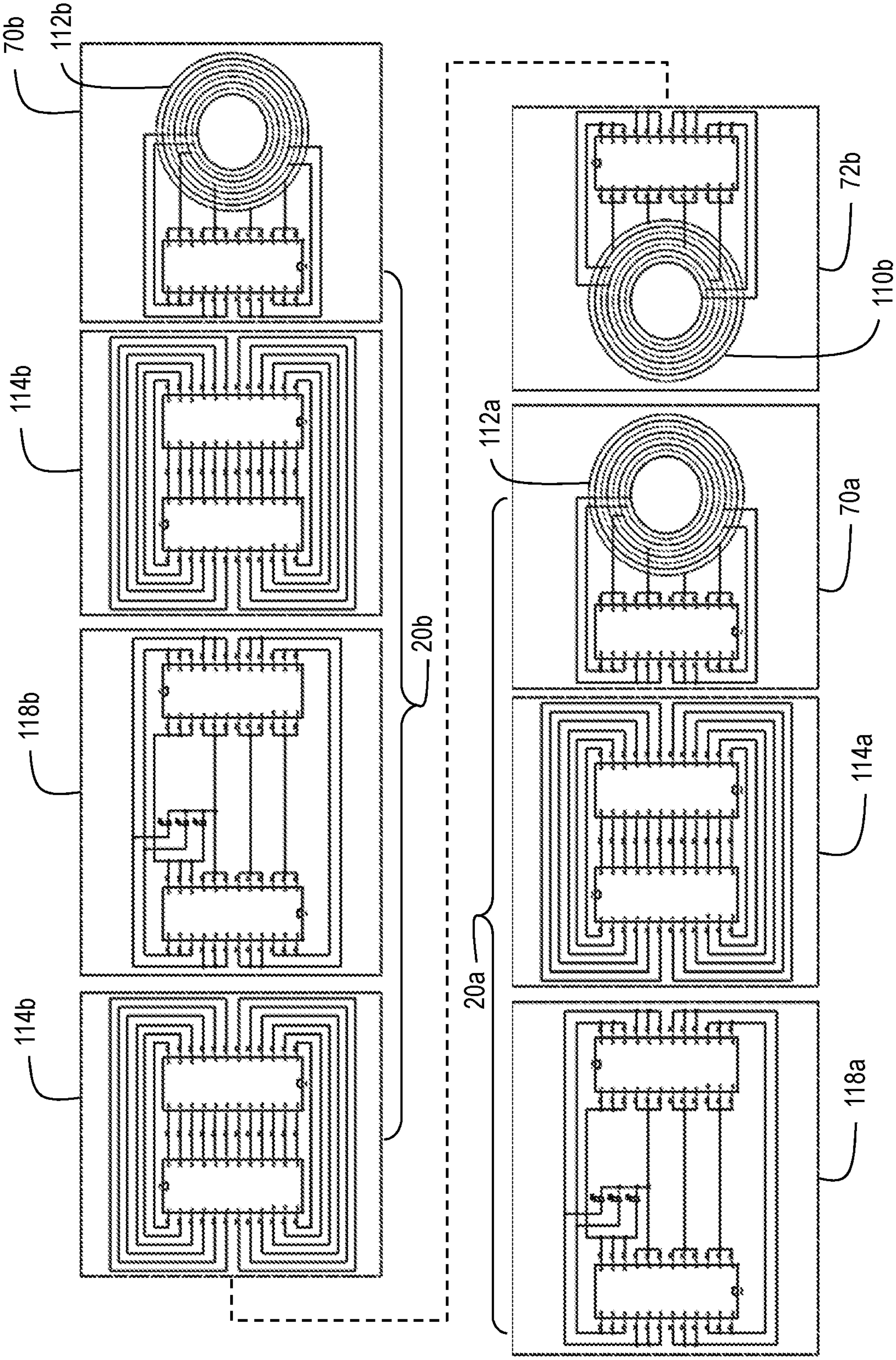


FIG. 11

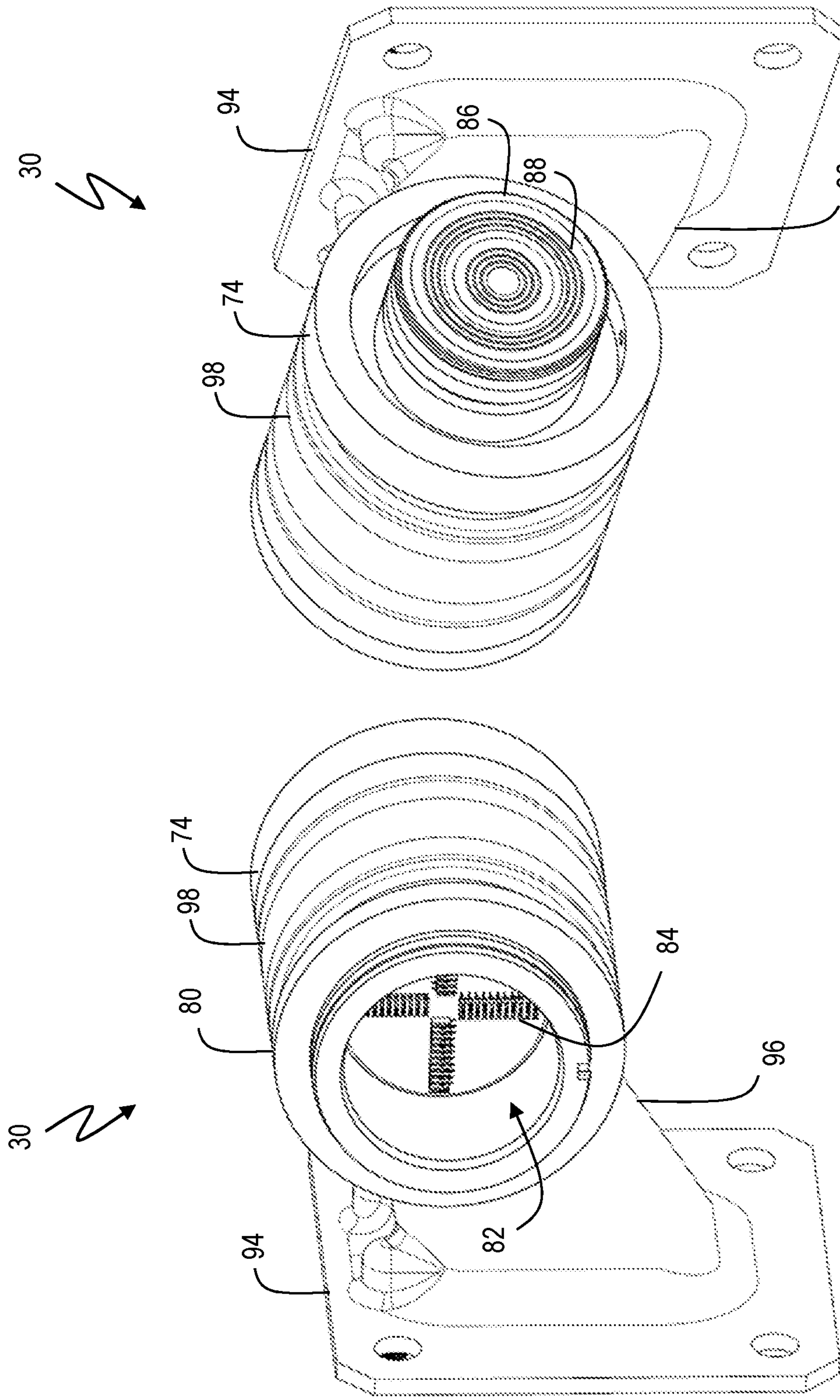


FIG. 12

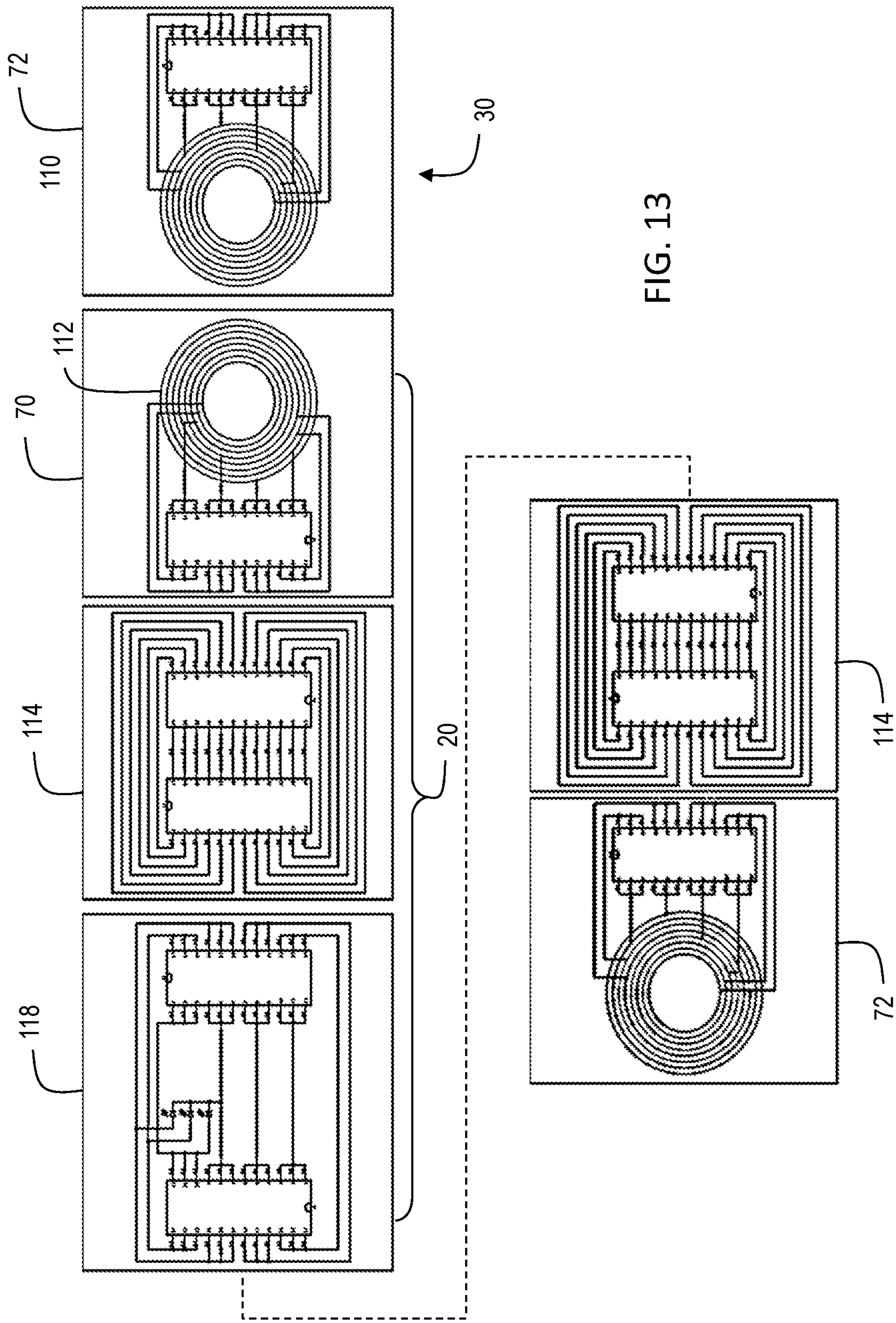


FIG. 13

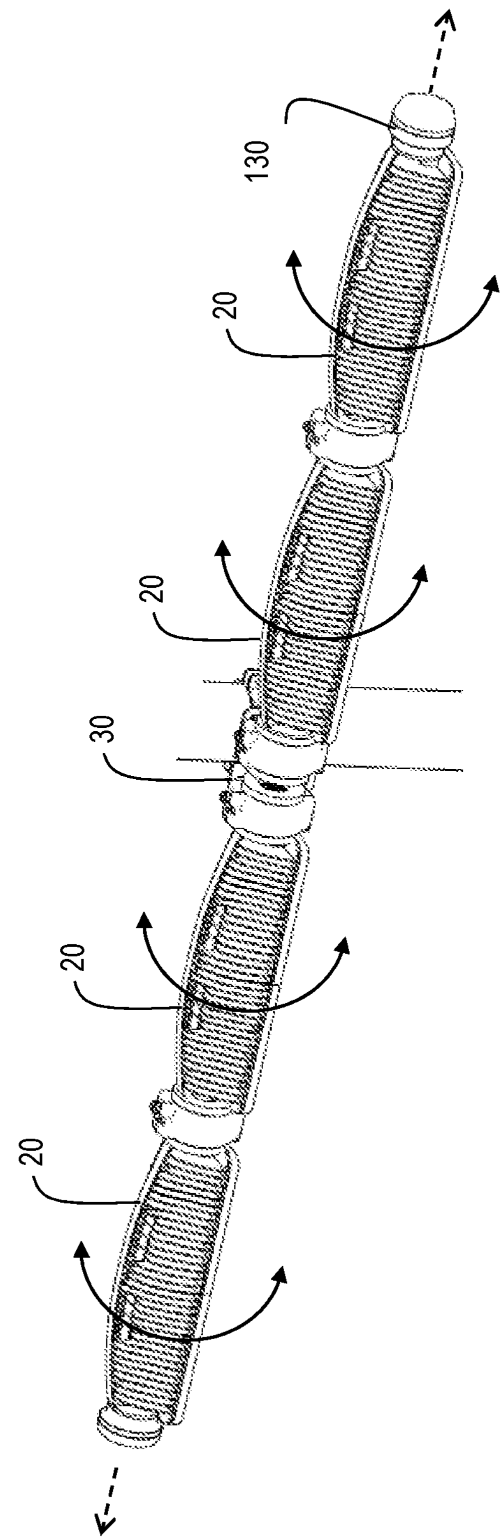


FIG. 14

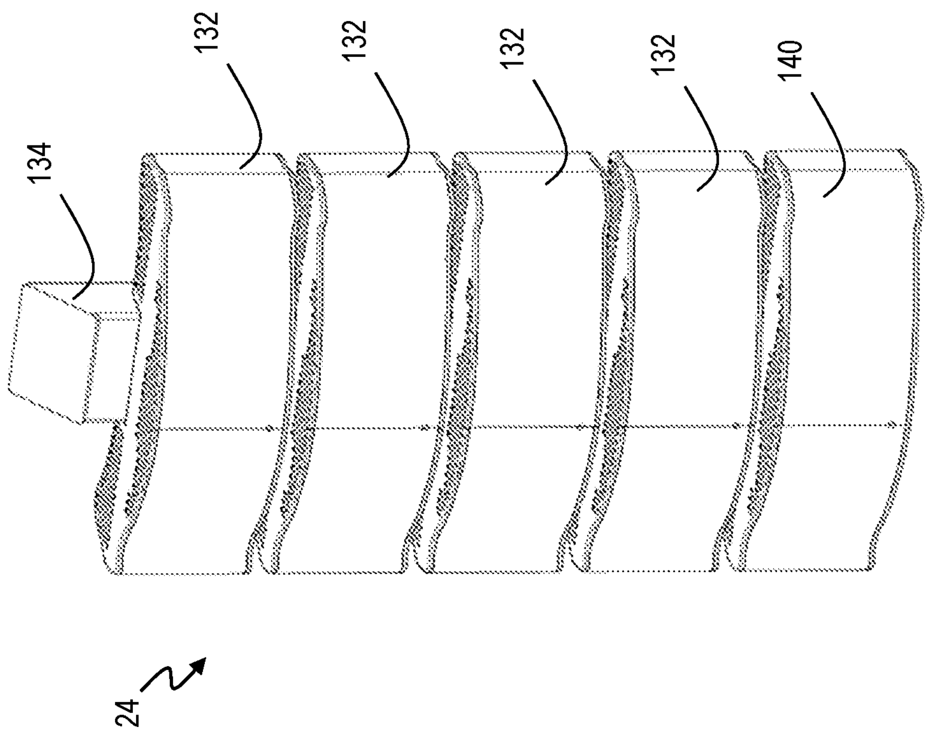


FIG. 15

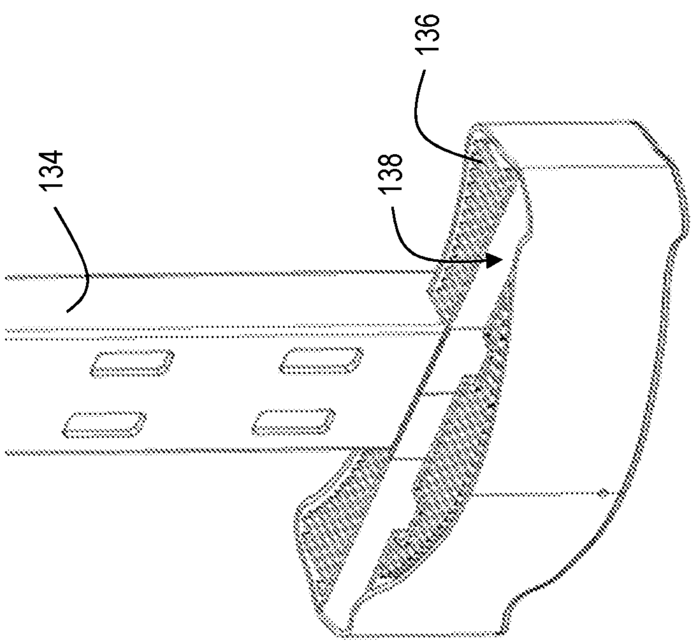


FIG. 16

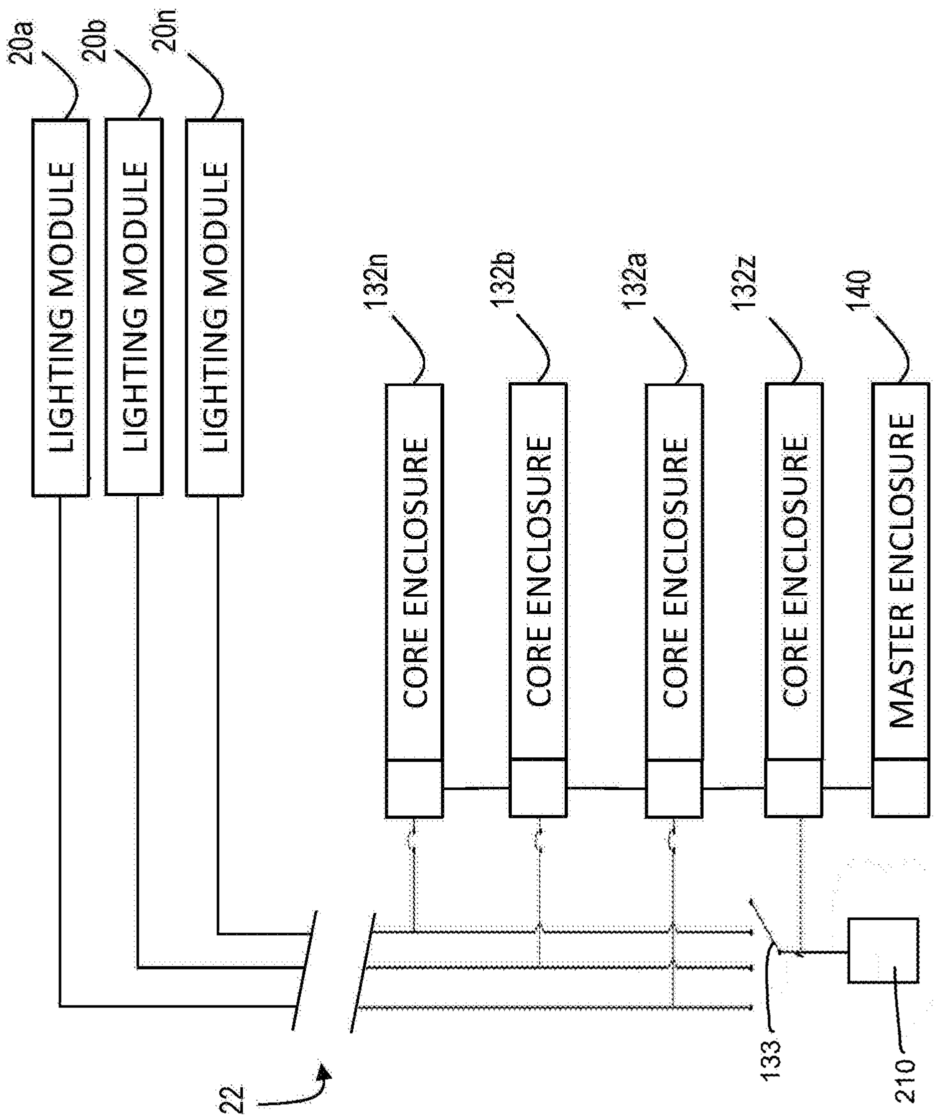

20a
20b
20n
LIGHTING MODULE
LIGHTING MODULE
LIGHTING MODULE
132n
132b
132a
132z
140
CORE ENCLOSURE
CORE ENCLOSURE
CORE ENCLOSURE
CORE ENCLOSURE
MASTER ENCLOSURE
22
133
210

FIG. 17

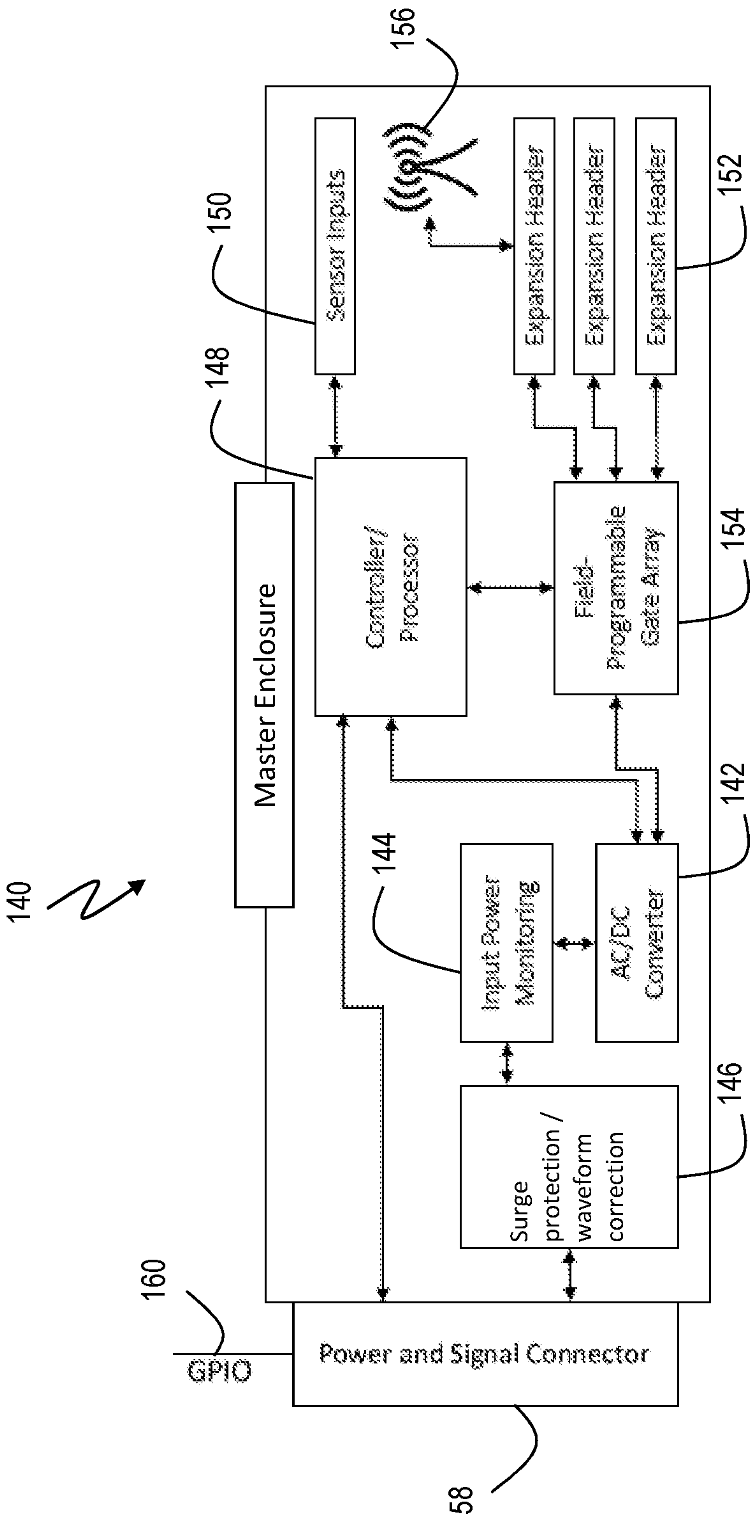

140
Master Enclosure
148
150
Sensor Inputs
156
Expansion Header
Expansion Header
Expansion Header
152
Controller/ Processor
Field-Programmable Gate Array
154
144
Input Power Monitoring
AC/DC Converter
142
Surge protection / waveform correction
146
160
GPIO
Power and Signal Connector
158

FIG. 18

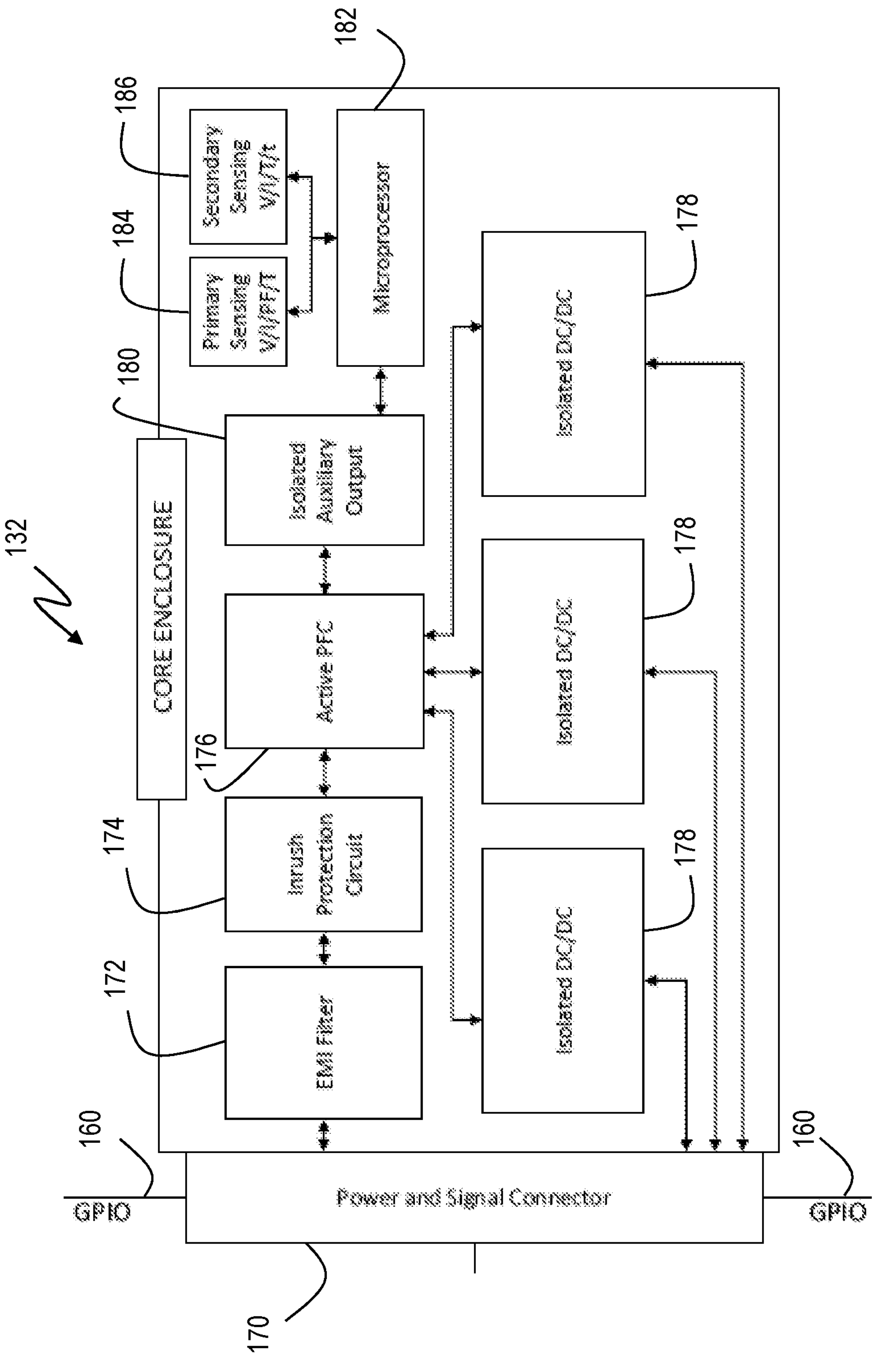

132
CORE ENCLOSURE
EMI Filter
Inrush Protection Circuit
Active PFC
Isolated Auxiliary Output
Primary Sensing V/I/PF/T
Secondary Sensing V/I/T/t
Microprocessor
Isolated DC/DC
Isolated DC/DC
Isolated DC/DC
Power and Signal Connector
GPIO
GPIO
160
160
170
172
174
176
178
178
178
180
182
184
186

FIG. 19

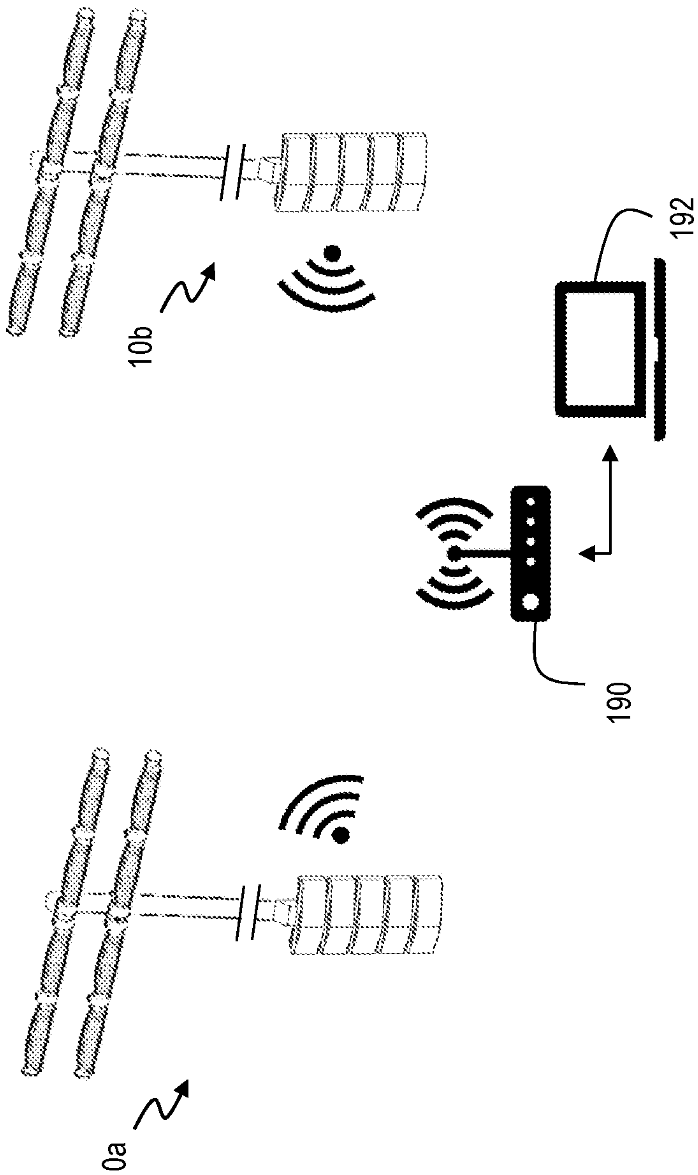


FIG. 20

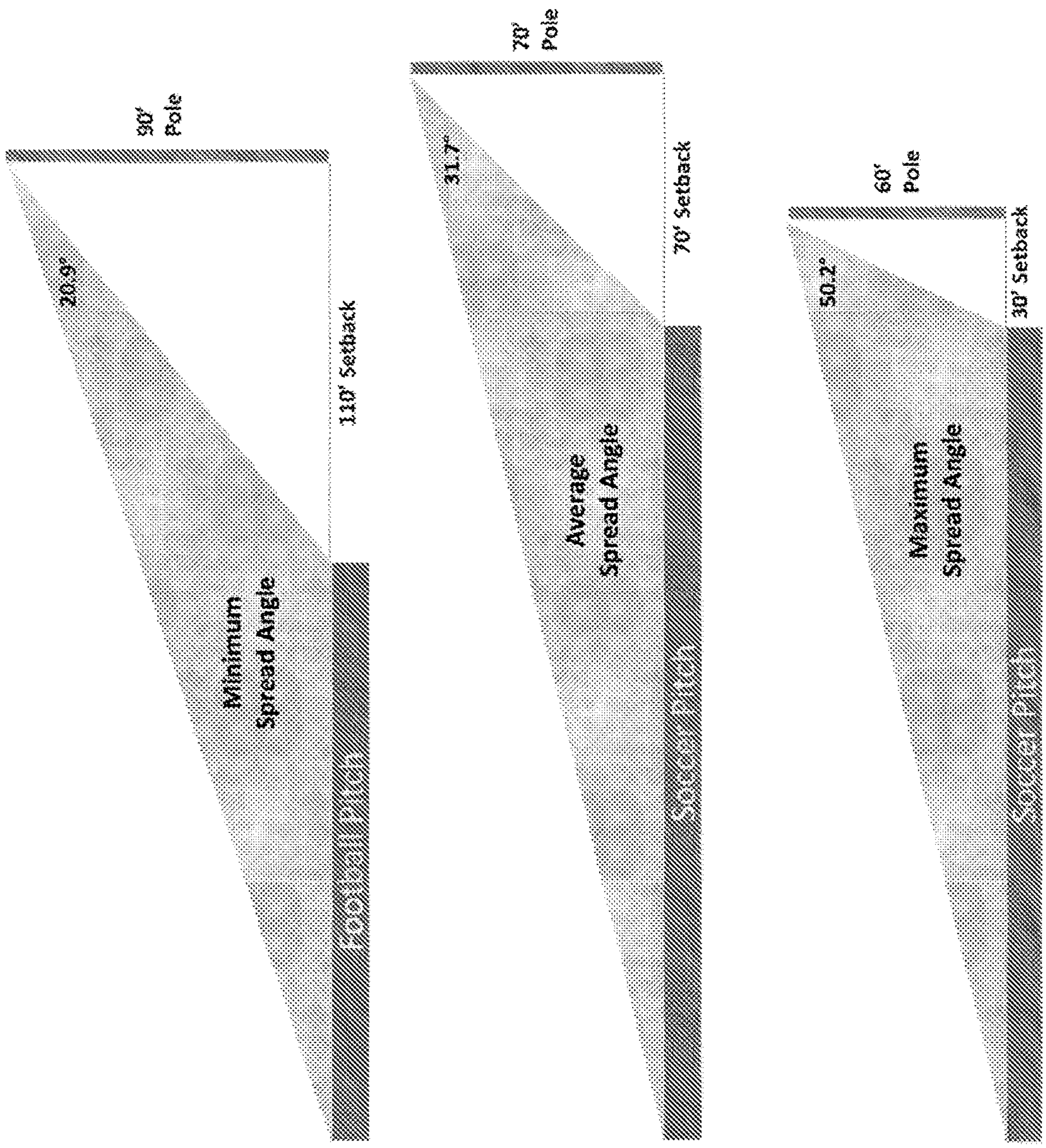

90' Pole
20.9°
110' Setback
Minimum Spread Angle
Football Pitch
70' Pole
31.7°
70' Setback
Average Spread Angle
Soccer Pitch
60' Pole
50.2°
30' Setback
Maximum Spread Angle
Soccer Pitch

FIG. 21

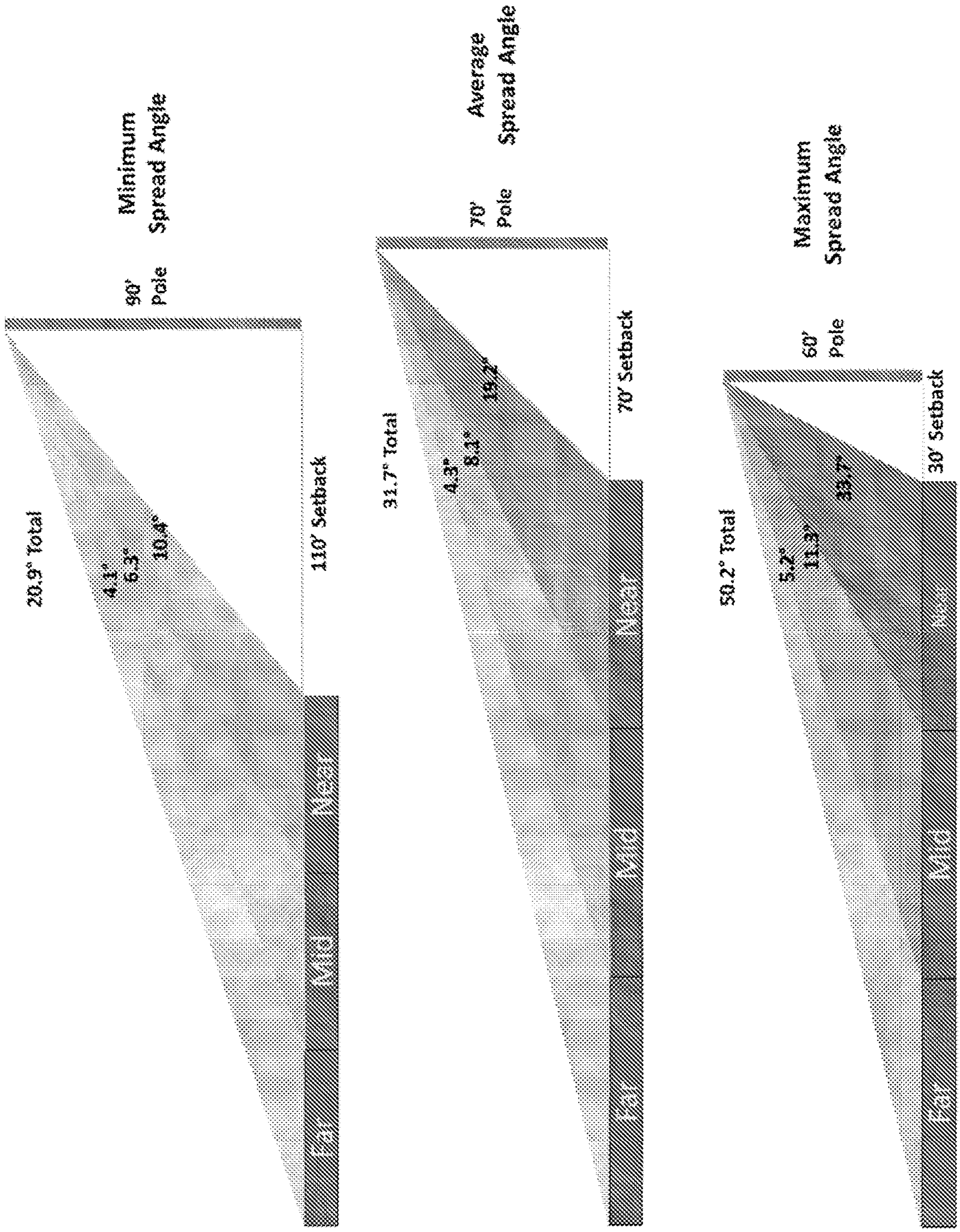

Minimum Spread Angle
90' Pole
110' Setback
20.9° Total
4.1°
6.3°
10.4°
Near
Mid
Far
Average Spread Angle
70' Pole
70' Setback
31.7° Total
4.3°
8.1°
19.2°
Near
Mid
Far
Maximum Spread Angle
60' Pole
30' Setback
50.2° Total
5.2°
11.3°
33.7°
Mid
Far

FIG. 22

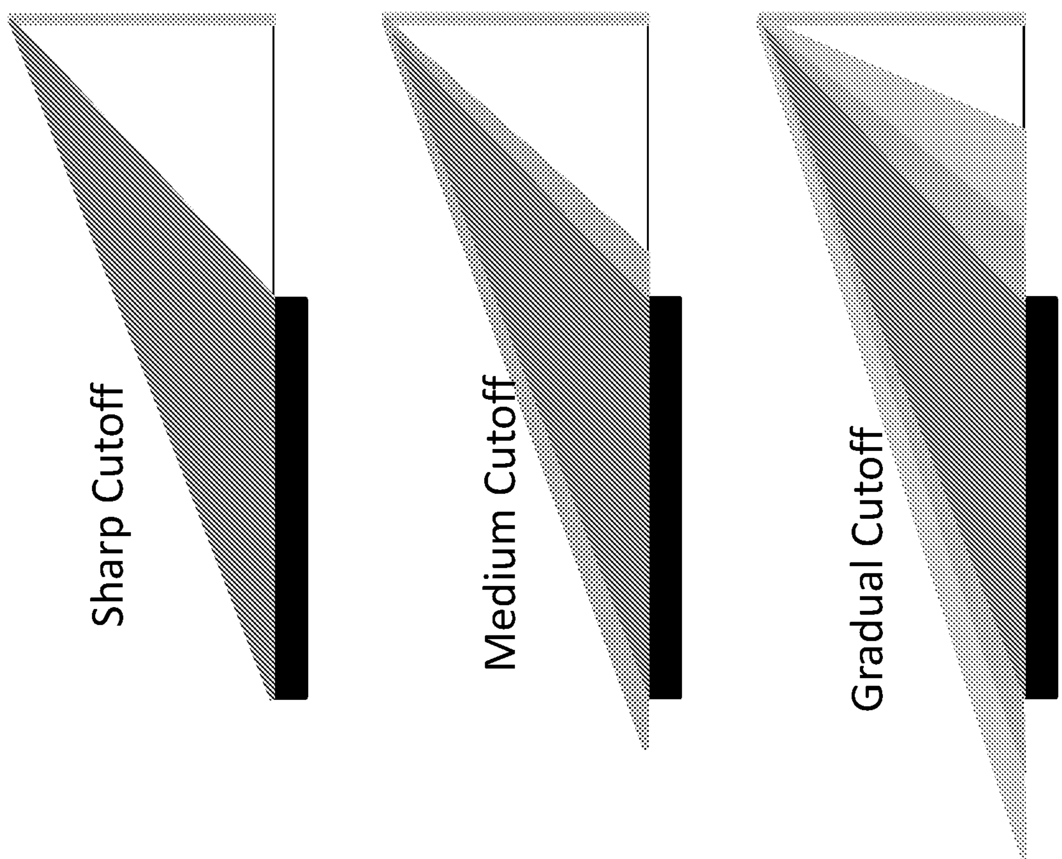


FIG. 23

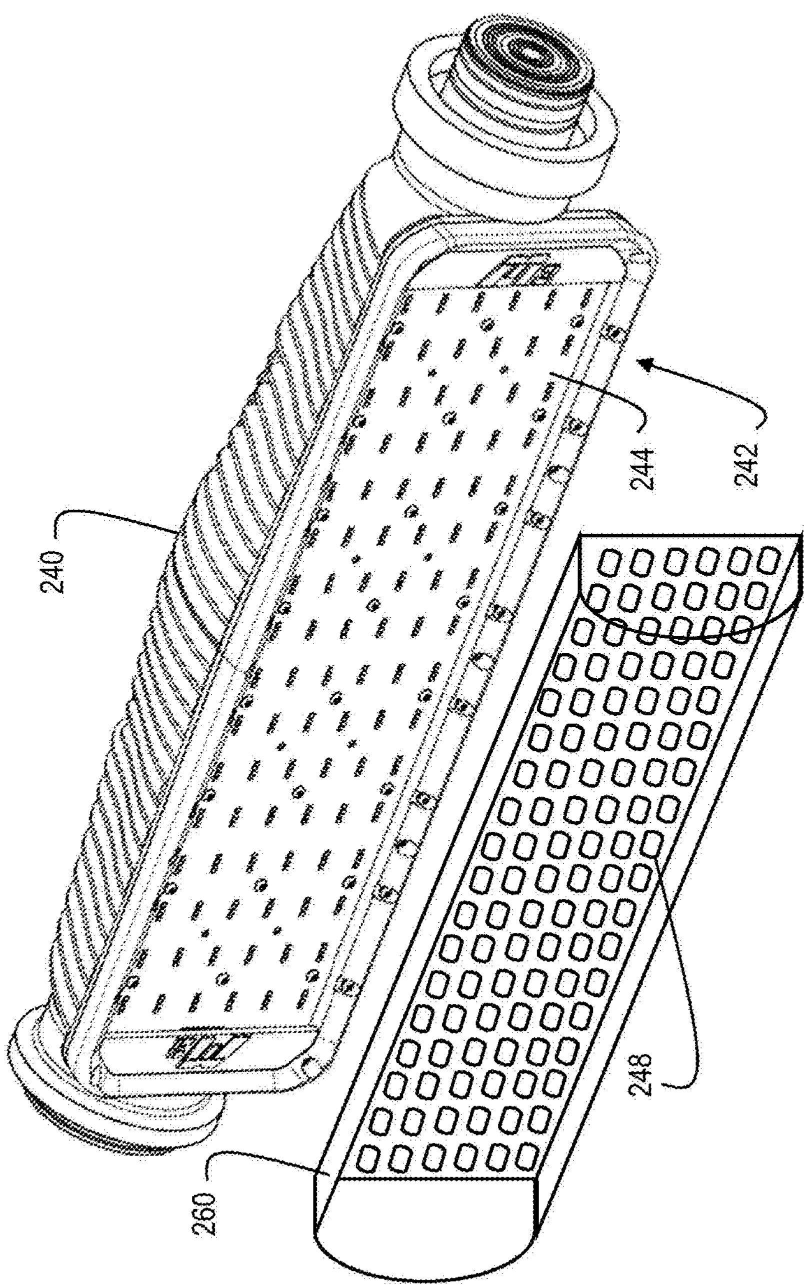


FIG. 24

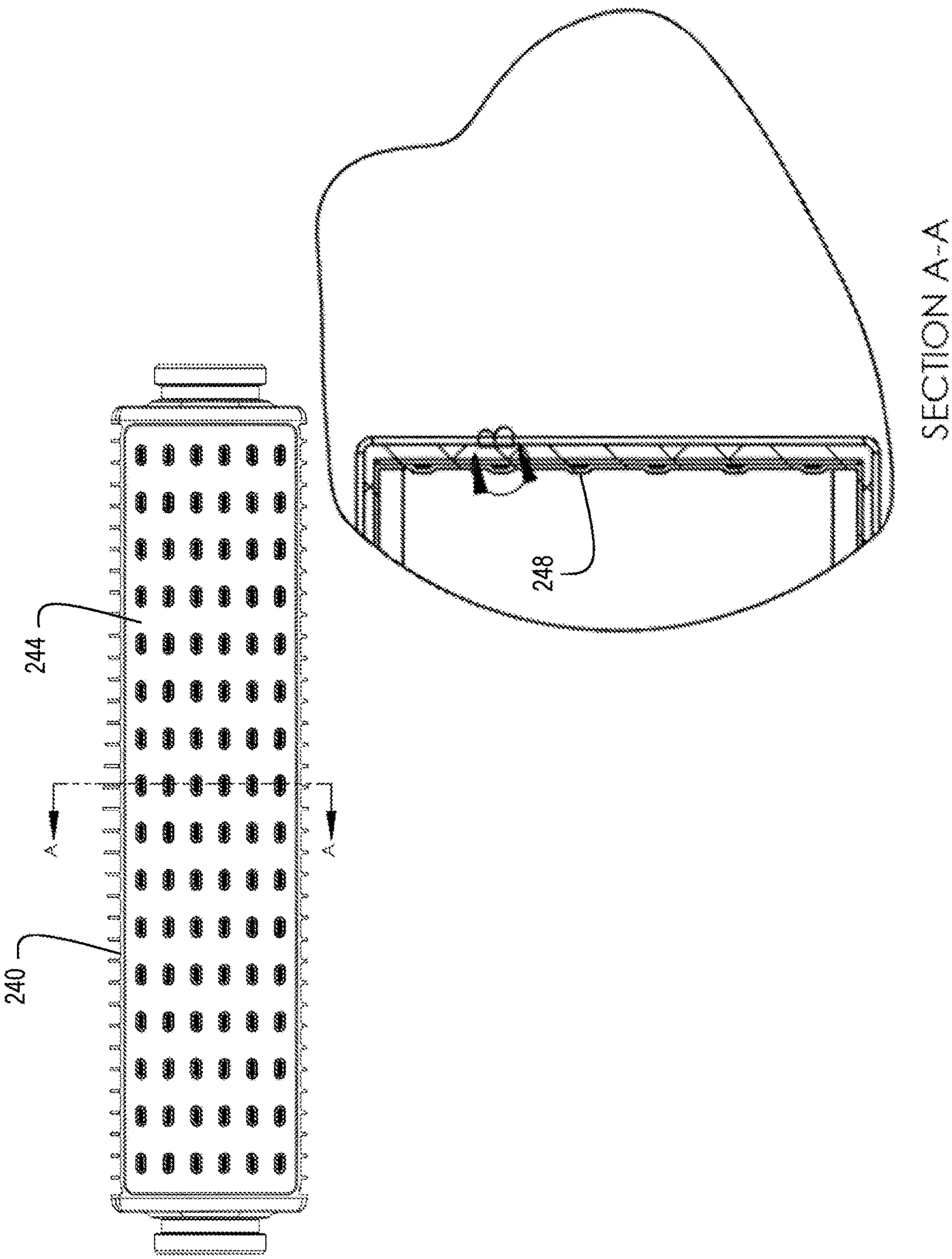


FIG. 25

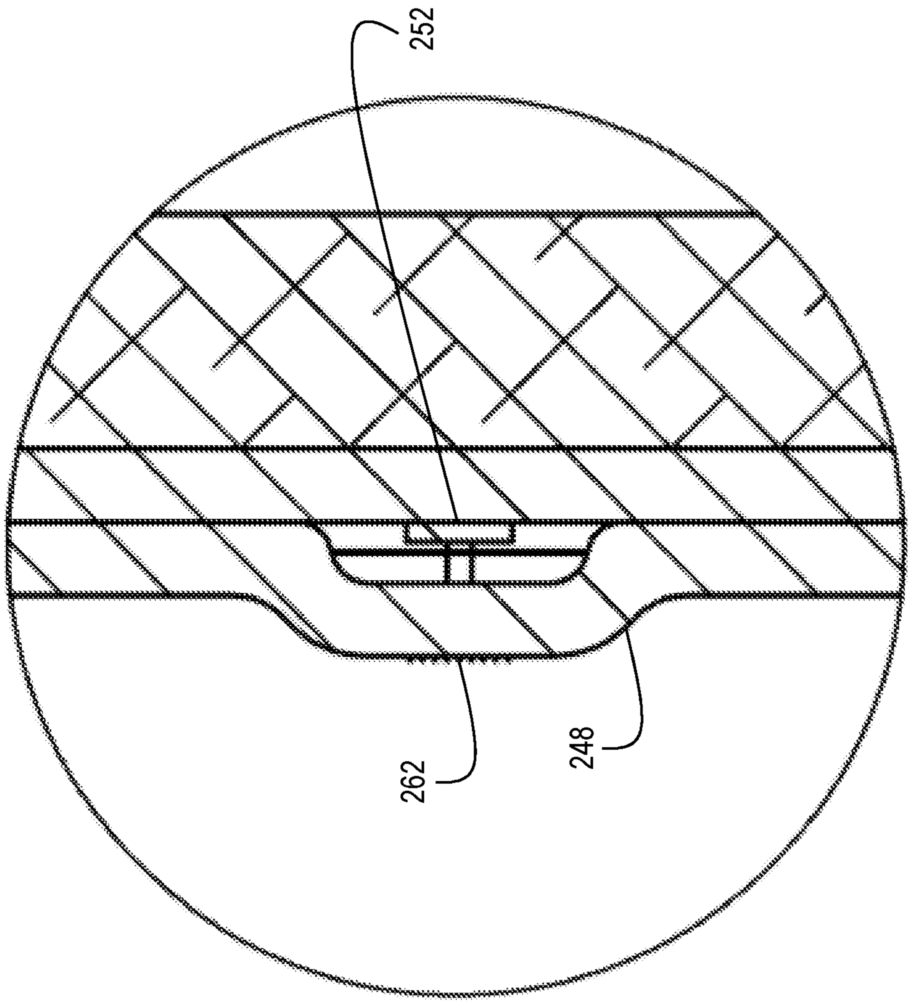


FIG. 26

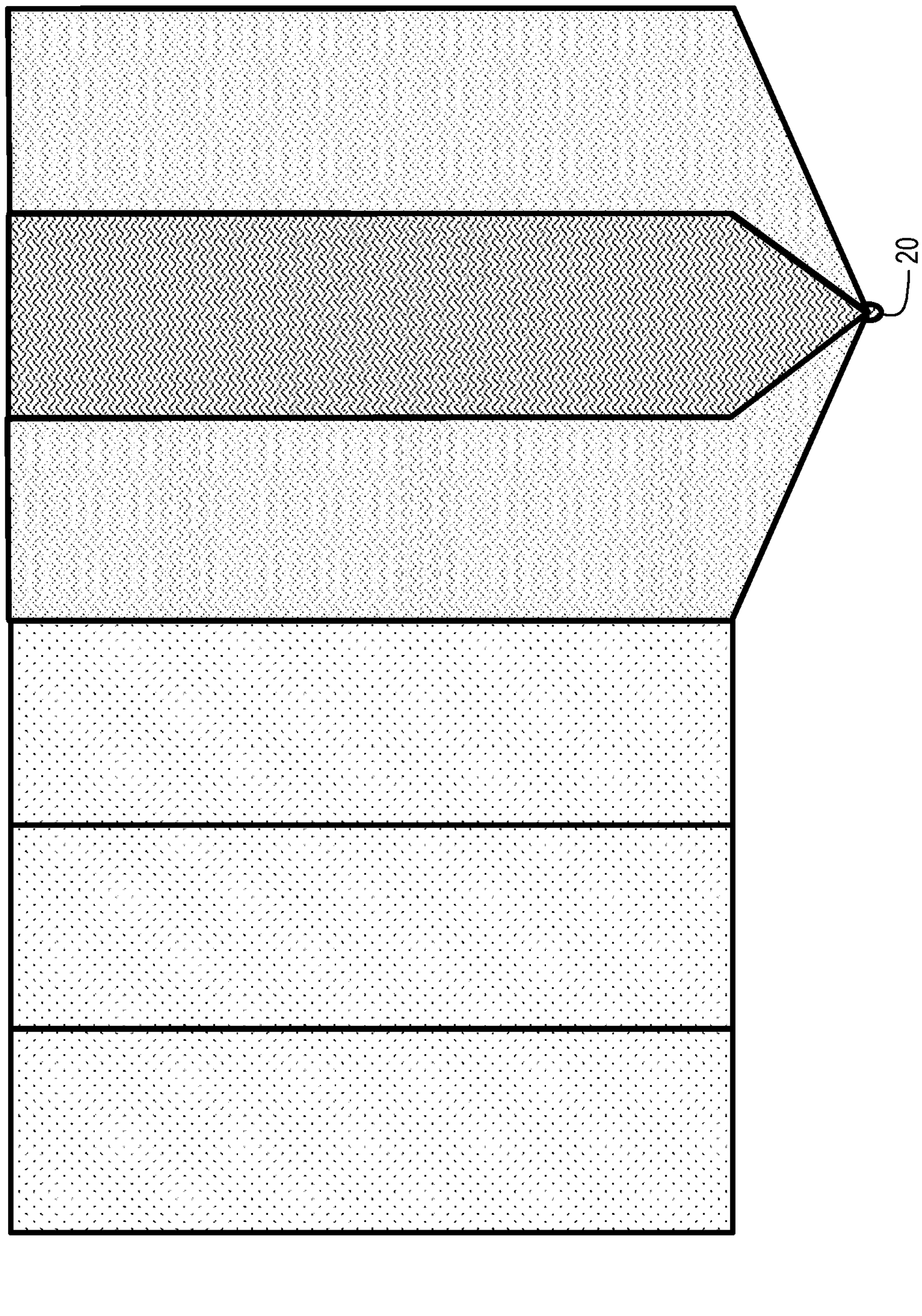


FIG. 27

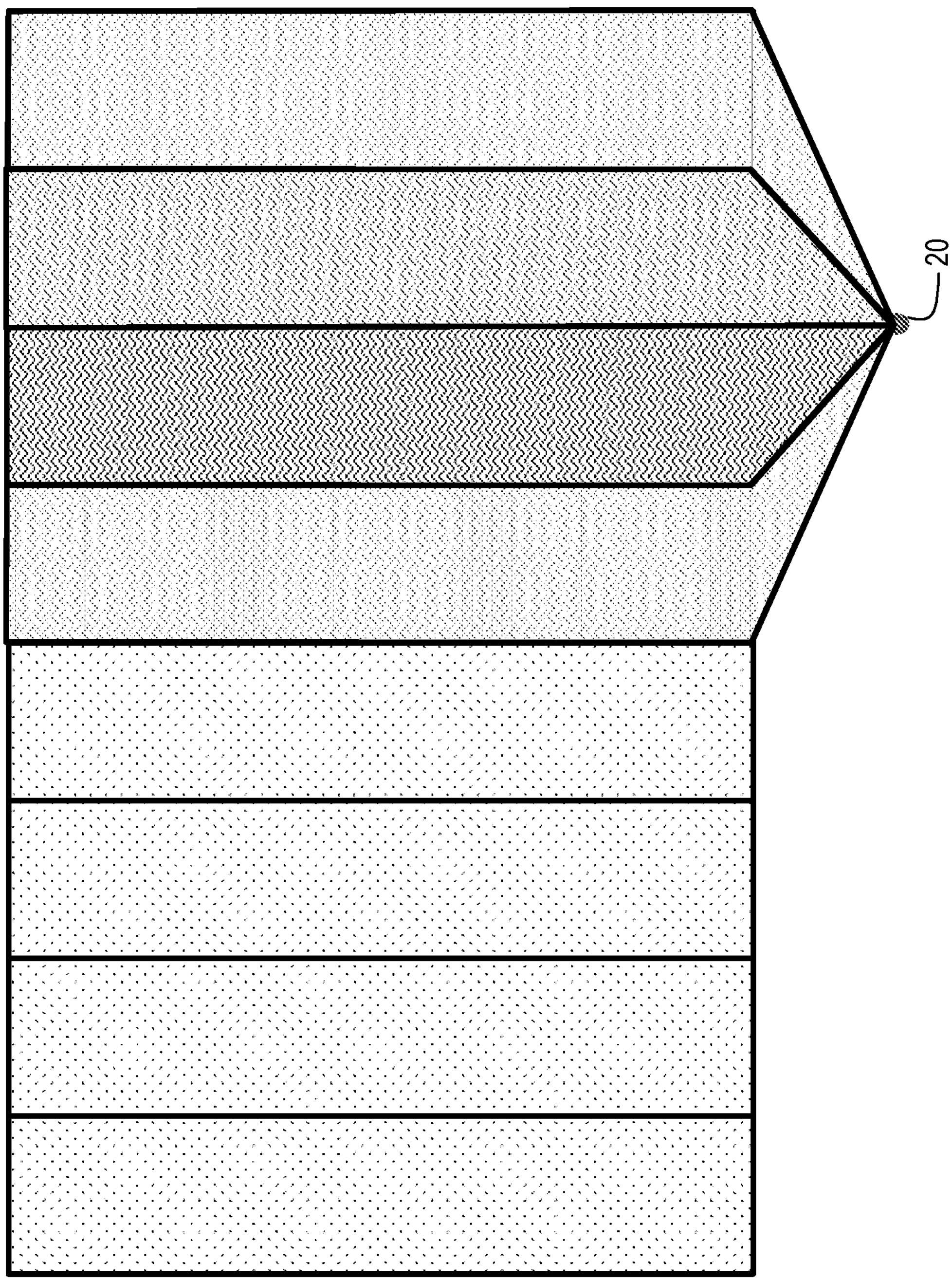


FIG. 28

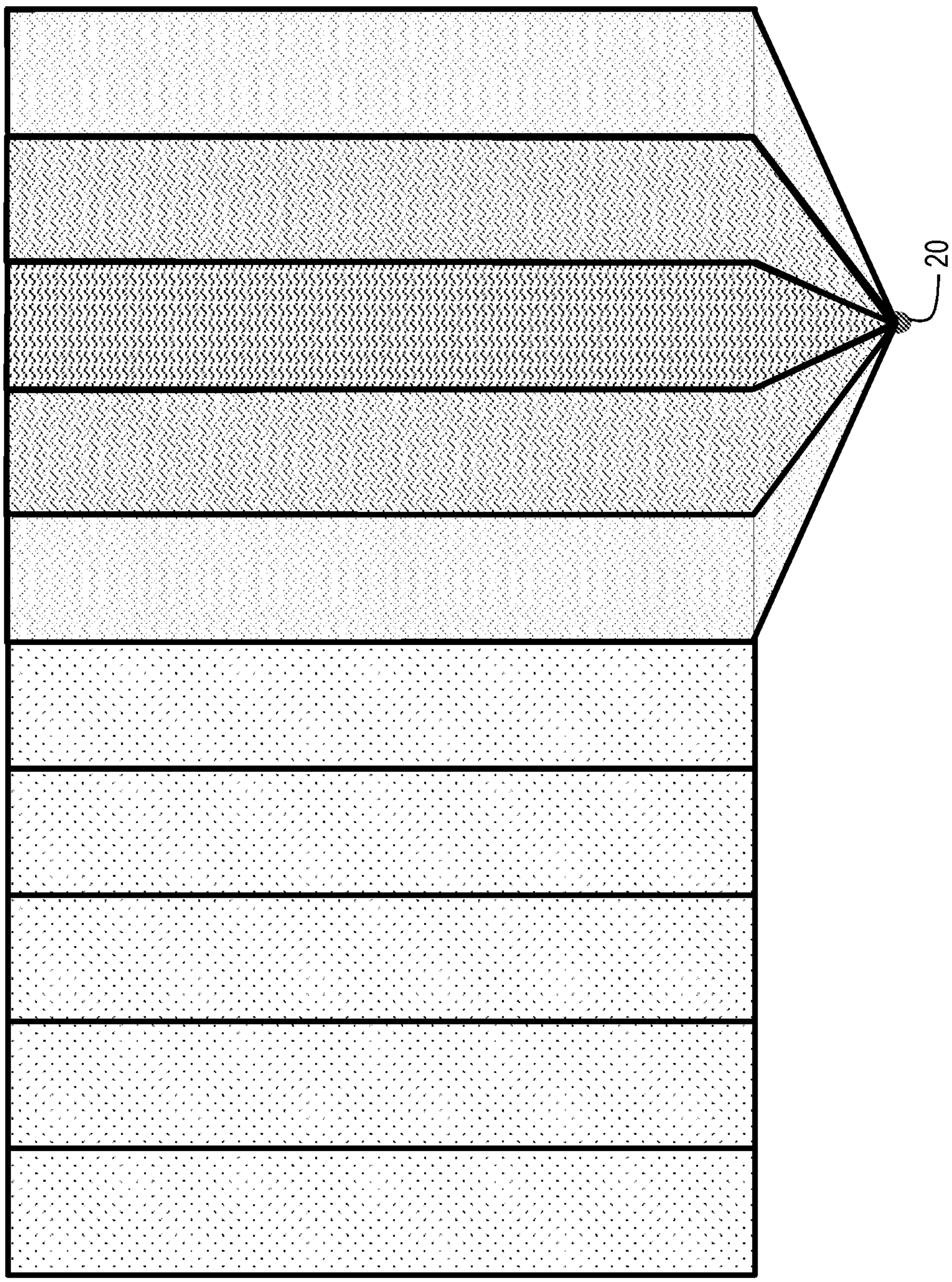


FIG. 29

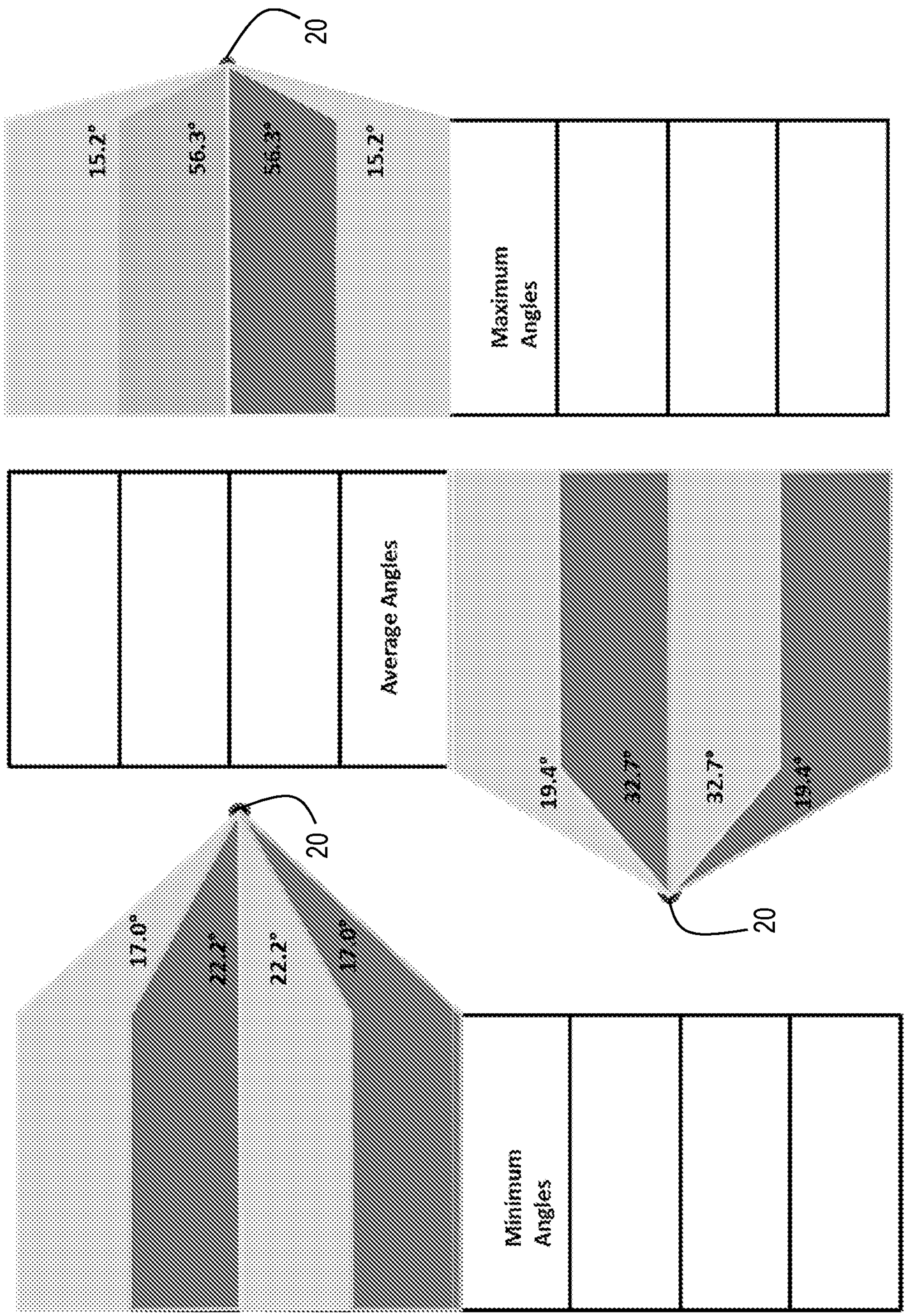

20
15.2°
56.3°
56.3°
15.2°
Maximum Angles
Average Angles
19.4°
32.7°
32.7°
19.4°
20
20
17.0°
22.2°
22.2°
17.0°
Minimum Angles

FIG. 30

ELECTRIC VEHICLE CHARGING STATION

[0001] This application claims priority under 35 U.S.C. § 119 to U.S. Provisional Application No. 63/559,120, filed Feb. 28, 2024, which is herein incorporated by reference in its entirety. This application incorporates U.S. application Ser. No. 18/913,748, filed Nov. 13, 2024, by reference in its entirety.

BACKGROUND OF THE INVENTION

Technical Field

[0002] The present invention relates to devices, systems, and methods for electrical vehicle charging stations and, more specifically, to electrical vehicle charging stations integrated with sports field and stadium lighting systems featuring modular auxiliary lighting systems employing asymmetric illumination sources.

Background Information

[0003] At sports fields and stadiums, lighting systems are employed to allow sports and other events to continue after sunset or indoors. Whether by accepted standards or law, in the event of an emergency where there is a loss of power to the main lighting systems, facilities are equipped with auxiliary or emergency lighting to allow the patrons of the facility to move to safety. Many municipal sports fields as well as elementary to high-school level venues fail to provide efficient and effective lighting systems in the event of a loss of power. When there is a loss of power to the main lighting system, a generator is often used to switch on to power a separate system of emergency lights. This system is slow to switch on, requires a lot of space for generators, is expensive, and can be noisy.

[0004] More efficient systems may provide emergency lighting below a threshold where the power source is a battery. With more efficient stadium lighting systems, once the emergency battery is charged, electricity demand may be low, compared to existing stadium lighting systems.

[0005] With increased concerns of global warming, there is diminished fossil fuel consumption and a shift to more renewable sources. With this push, people are also attempting to make energy consumption more efficient, and manufacturers are making and selling more cars powered by electricity. Despite both a government and a public push, the infrastructure for charging electric cars is currently lacking.

SUMMARY OF THE INVENTION

[0006] The present disclosure is directed toward systems, methods, and devices, providing electrical vehicle charging.

[0007] In one aspect of the present disclosure provided herein, is an electrical vehicle charging system having a lighting system having a controller stack connected to a lighting module, an auxiliary lighting module by wiring, an alternating current power supply, a switching circuit, a local power storage device, and a wireless communication gateway. The electrical vehicle charging system further having a plurality of electrical vehicle chargers each having an electrical vehicle charging cable configured for connection to an electrical vehicle; a connection to an existing power grid; an internet of things module configured for two-way, wireless communication with the wireless gateway; a feedback system; and a control system connected to and configured for communication with the internet of things module, the existing power grid, and the electrical vehicle charging cable.

[0008] In one aspect of the present disclosure provided herein, is a lighting system, having at least one lighting module, the lighting module having a housing extending from first end to a second end along a longitudinal axis and having an elongated opening in alignment with an illumination source therein; a first coupler positioned at the first end of the housing and has an end face having a first set of electrical contacts and a first set of auxiliary electrical contacts; and a second coupler positioned at the second end of the housing and an internal bore having a second set of electrical contacts and a second set of auxiliary electrical contacts within the internal bore. The lighting system further has an auxiliary lighting module, having a housing having an opening and an illumination source therein, and a coupler connected to the auxiliary lighting module housing having a set of auxiliary module electrical contacts and the coupler configured for connection to one of the first coupler or the second coupler; and a local power storage device. The lighting system further has a controller stack connected to the at least one lighting module by a first set of wiring, and connected to the auxiliary lighting module by a second set of wiring, the controller stack having: an alternating current power supply; a wireless gateway; and a switching circuit having a voltage monitor connecting the alternating current power supply and the first set of wiring, and the local power storage device and the second set of wiring, the voltage monitor measuring for a voltage drop threshold. The first set of electrical contacts are connected to the second set of electrical contacts internally of the lighting module housing, and the second set of wiring connects the first set of auxiliary electrical contacts and the second set of auxiliary electrical contacts. The lighting system further has a plurality of electrical vehicle chargers each connected to the controller stack having: an electrical vehicle charging cable configured for connection to an electrical vehicle; a connection to an existing power grid; an internet of things module configured for two-way, wireless communication with a wireless gateway; a feedback system; a control system connected to and configured for communication with the internet of things module, the existing power grid, and the electrical vehicle charging cable.

[0009] These, and other objects, features and advantages of this invention will become apparent from the following detailed description of the various aspects of the invention taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The present invention will be more fully understood and appreciated by reading the following Detailed Description in conjunction with the accompanying drawings, in which:

[0011] FIG. **1** is a perspective view of an asymmetric source sports lighting system in accordance with an aspect of the present invention;

[0012] FIG. **2** is a perspective view of the upper portion of a support pole of an asymmetric source sports lighting system in accordance with an aspect of the present invention;

[0013] FIG. **3** is a perspective view of the asymmetric lighting source for a lighting module in accordance with an aspect of the present invention;

[0014] FIG. **4** is a mechanical view of the light emitting diode (LED) layout for an asymmetric lighting source in accordance with an aspect of the present invention;

[0015] FIG. **5** is schematic of the electronics for an asymmetric lighting source in accordance with an aspect of the present invention;

[0016] FIG. **6** is a perspective view of a lighting module in accordance with an aspect of the present invention having a lens array thereon;

[0017] FIG. **7** is a perspective view of the male and female couplers of a lighting module in accordance with an aspect of the present invention;

[0018] FIG. **8** is a cross-sectional view of the male and female couplers of a lighting module in accordance with an aspect of the present invention;

[0019] FIG. **9** is a perspective view of a coupler clamp for securing lighting modules to each other in accordance with an aspect of the present invention

[0020] FIG. **10** is cross-sectional view of a lighting module to lighting module connection in accordance with an aspect of the present invention;

[0021] FIG. **11** is an electrical diagram of a lighting module to lighting module connection in accordance with an aspect of the present invention;

[0022] FIG. **12** is two perspective views of a mount in accordance with an aspect of the present invention;

[0023] FIG. **13** is an electrical diagram of a lighting module to mount connection in accordance with an aspect of the present invention;

[0024] FIG. **14** is a perspective view showing axial rotation of a series of interconnected lighting modules in accordance with an aspect of the present invention;

[0025] FIG. **15** is a perspective view of a controller stack in accordance with an aspect of the present invention;

[0026] FIG. **16** is a perspective view of a core enclosure in accordance with an aspect of the present invention;

[0027] FIG. **17** is high level schematic for a lighting system in accordance with an aspect of the present invention;

[0028] FIG. **18** is a detailed schematic of a master controller in accordance with an aspect of the present invention;

[0029] FIG. **19** is a detailed schematic of a core enclosure in accordance with an aspect of the present invention

[0030] FIG. **20** is a schematic of wireless monitoring and control approach in accordance with an aspect of the present invention; and

[0031] FIG. **21** is a schematic of beam steering using a lighting system in accordance with an aspect of the present invention;

[0032] FIG. **22** is a schematic of beam angles changes using a lighting system in accordance with an aspect of the present invention;

[0033] FIG. **23** is a schematic of tunable cut-off in a lighting system in accordance with an aspect of the present invention;

[0034] FIG. **24** is a perspective view of an environmental sealing system for a lighting module in accordance with an aspect of the present invention;

[0035] FIG. **25** is a front view of an environmental sealing system for a lighting module in accordance with an aspect of the present invention;

[0036] FIG. **26** is a side view of a micro-lens for a lighting module in accordance with an aspect of the present invention;

[0037] FIG. **27** is a first view of illumination steering using a lens array in accordance with an aspect of the present invention;

[0038] FIG. **28** is a second view of illumination steering using a lens array in accordance with an aspect of the present invention;

[0039] FIG. **29** is a third view of illumination steering using a lens array in accordance with an aspect of the present invention; and

[0040] FIG. **30** is a fourth view of illumination steering using a lens array in accordance with an aspect of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0041] The present invention will be discussed in detail in terms of various exemplary embodiments according to the present invention with reference to the accompanying drawings. In the following detailed description, numerous specific details are set forth to provide a thorough understanding of the present invention. To those skilled in the art, it will be obvious that the present invention may be practiced without these specific details. Similarly, well-known structures are not described to avoid obscuring the present invention.

[0042] Thus, the implementations described below are exemplary implementations provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims.

[0043] Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, summary, or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the claims. Specific dimensions and other physical characteristics relating to the embodiments disclosed are not to be considered as limiting, unless the claims expressly state so.

[0044] Likewise, the various figures, steps, procedures, and workflows are presented only as an example and in no way limit the systems, methods, or apparatuses described to perform their respective tasks or outcomes in different timeframes or orders. Unless expressly stated, any method set forth herein shall not be construed as requiring that its steps be performed in a specific order. The teachings of the present invention may be applied to any asymmetric lighting system that has or is integrated with an auxiliary lighting system.

[0045] The various embodiments described herein provide for systems, devices, and methods for asymmetric lighting systems that have or are integrated with auxiliary lighting systems: particularly, for lighting systems for sports and auxiliary lighting systems for asymmetric source sports lighting systems.

[0046] Referring to the figures, wherein like numerals refer to like parts throughout, there is seen in FIG. **1** an asymmetric source sports lighting system **10** according to the present invention. System **10** is designed for installation on a support pole **12** to provide illumination over a target area **14**, such as a sporting field or pitch. The system **10** may also be used for lighting in outdoor parking areas, municipal street lighting, roadway lighting, or at any place where similar outdoor lighting systems are used. System **10** includes one or more rows of light emitting diode (LED) lighting modules **20** that extend laterally from support pole **12**. Lighting modules **20** are powered via a wiring harness **22** that extends along the interior of support pole **12** and is coupled to a controller stack **24**. Controller stack **24** transforms local building power from AC to DC and includes LED drivers **26** for lighting modules **20**. A battery **210** is connected to support pole **12**. The battery **210** may be directly connected to the controller stack **24**, or the battery may be a separate unit connected by internal wiring to the controller stack **24**. The controller stack **24** charges the battery **210** from transformed local building power while AC power is received by a power supply. In the event of black-out or brown-out conditions or a drop below a threshold voltage, the power supply is switched to battery power the lighting modules **20** or a subset of lighting modules. While a battery is described, a local electrical power storage of other kinds may be used in place of a battery.

[0047] In certain other embodiments, lighting modules **20** may include one or more columns of laser diodes (LD) instead of light emitting diodes. Controller stack **24**, may also include LD drivers for lighting modules **20**. The lighting systems described herein are LED based systems. However, the invention described could be used with LD based systems as well.

[0048] Referring to FIG. **2**, a central mount **30** is coupled to pole **12** and used to support first and second lighting modules **20**. Lighting modules **20** are coupled to either side of mount **30** using a modular coupling system described herein that physically supports modules **20** and electronically interconnects modules **20** to wiring harness **22** and thus controller stack **24**. The opposing end of each lighting module **20** coupled to mount **30** may be used to physically support and electronically interconnect to additional lighting modules **20** extending further outwardly from support pole **12**. The combination of lighting modules **20** connected to mount **30** and the additional lighting modules **20** extending to either side of pole **12** are self-supporting so that support pole **12** does not need to include physical cross-arms or lateral supports to mount additional lighting modules **20**. The particular dimensions of lighting module **20** may be varied as desired. For example, lighting module **20** could be provided in two lengths, X and **2**X, that may be mixed and matches as needed for a particular installation.

[0049] Continuing with reference to FIG. **2**, two sets of lighting modules are depicted. The second set of lighting modules may be, for example, auxiliary emergency lighting **200**. Auxiliary emergency lighting **200** may be turned off during normal lighting conditions and activated when emergency conditions arise. In other embodiments, auxiliary emergency lighting **200** may operate during normal lighting conditions but remain activated during emergency conditions.

[0050] Referring to FIG. **3**, each lighting module **20** includes a housing **40** extending along a longitudinal axis X-X. Housing **40** defines a rectangular opening **42** in a central portion thereof that permits access to an asymmetric illumination source **44**. Asymmetric illumination source **44** is dimensioned to produce a rectangular beam of illumination from rectangular opening **42** of module **20**. Housing **40** may further include fins **46** or other external structures for dispersing heat generated by using asymmetric illumination source **44**.

[0051] Referring to FIGS. **4** and **5**, asymmetric illumination source **44** comprises multiple rows **50** of light emitting diode (LED) sets **52** spaced along a substrate **54** and coupled to electronic circuitry **56** for asymmetrically driving illumination source **44**. Each row **50**, or optionally, each pair of rows **50**, are independently controllable by adjusting the amount of power delivered to that row (or pair or rows) using electronic circuitry **56** and controller stack **24** to provide asymmetric illumination from module **20**. Optionally, a local microcontroller in each module **20** can be for further adjustment of the amount of power provided to each row (or pair or rows) of LED sets. As seen in FIG. **5**, asymmetric illumination source **44** having three independently controllable rows **50** of LED sets **52**. Electronic circuitry **56** further includes pass-through circuitry **58** for providing power to adjacently connected lighting modules **20** that also include independently controlled rows **50** of LED sets **52**. In the example of FIG. **5**, a total of two additional lighting modules **20** may be interconnected and supported by circuitry **58**.

[0052] Referring to FIG. **6**, a molded lens array **60** is positioned over an asymmetric illumination source **44** to reduce harshness and provide sealing of asymmetric illumination source **44** within housing **40**. Housing **40** of module **20** is further configured to allow for easy coupling to the support pole and to other housings **40**, forming both structural and electrical connection. Housing **40** includes a male coupler **70** positioned at one end of housing **40** and a female coupler **72** positioned at an opposing end of housing **40**. Male coupler **70** is defined by a radially extending flange **74** and a circumferentially extending, outwardly facing bearing surface **76**. Female coupler **72** includes a correspondingly dimensioned flange **78** and a receptacle **82** defining a circumferentially extending, inwardly facing bearing surface **77**.

[0053] Referring to FIGS. **7** and **8**, female coupler **72** further includes a set of brush contacts **84** positioned in receptacle **82** that face outwardly along axis X-X and male coupler **70** includes an end face **86** supporting set of ring contacts **88** that face outwardly in the opposite direction along axis X-X from brush contacts **84**. Male coupler **70** may additionally include grooves **90** formed therein to house an O-ring for sealing purposes. It should be recognized that other contacts may be used, such as pogo pins and the like. As detailed below, brush contacts **84** and ring contacts **88** define a plurality of independent pathways for powering the independently controlled rows **50** of LED sets **52**.

[0054] Referring to FIGS. **9** and **10**, a clamp **92** may be positioned and secured in covering relation to flanges **74** and **80** to secure a first module **20** *a* to a second module **20** *b* when male coupler **70** and female coupler **72** are full joined so that bearing surfaces **76** and **77** are in seated together and brush contacts **84** and ring contacts **88** are in contact and electrically engaged. Clamp **92** comprises a pair of jaws **100** and **102** that can be opened and then closed in covering relation to flanges **74** and **80**, as seen in FIG. **10**, when male coupler **70** of one module **20** *a* is jointed with and seated inside female coupler **72** of an adjacent module **20** *b*. When male coupler **70** is fully inserted into female coupler **72**, flanges **74** and **80** will abut and brush contacts **84** will physically and electrically engage ring contacts **88**. Clamp **92** may then be closed over flanges **74** and **80** to secure first module **20** *a* to second module **20** *b* using a latch **104** on one jaw **102** that cooperates with a slot **106** in the other jaw **100**, with electrical continuity between first module **20** *a* to second module **20** *b* provided via the engagement of ring contacts **88** with brush contacts **84**. Adjacent modules **20** may thus be electrically interconnected when coupled together so that each module **20** has multiple independent electrical power pathways for driving the independently controllable LED rows of asymmetric illumination source **44**.

[0055] Referring to FIG. **11**, module **20** *b* is electrically interconnected to module **20** *a* so that LED circuitry **118** *b* of module **20** *b* and LED circuitry **118** *a* of module **20** *a* are coupled together and under common power control. For example, coupler **70** *b* of module **20** *b* includes coupler circuitry **112** *b* that can receive power from ring contacts **88**. Coupler circuitry **112** *b* is coupled to LED circuitry **118** *b* via cabling **114** *b*. LED circuitry **118** *b* is also coupled to coupler circuitry **110** *b* associated with female coupler **72** *b* via cabling **114** *b*. As a result, independent power pathways for LED circuitry **118** *b* extend through module **20** *b* and are available at coupler **70** *b* and coupler **72** *b* such as that a power supply connected to coupler **70** will also provide power to coupler **72**, and vice versa. As further seen in FIG. **11**, module **20** *a* can be electrically coupled to module **20** *b* via a coupler **70** *a* that is secured to coupler **72** *b*. Coupler circuitry **112** *a* of module **20** *a* is coupled to LED circuitry **114** *a* via cabling **114** *a*. Although not illustrated for simplicity, it should be evident that module **20** also includes a coupler **72** *a* that can be, in turn, coupled to another module **20**, and so on, with the power supply for all housings **20** connected to an available coupler **70** or **72** at either end. Thus, module **20** is bi-directional and can be placed in series with additional housings **20** for common power control.

[0056] Referring to FIG. **12**, mount **30** for attaching one or more housings **20** to a support pole **12** comprises a mounting plate **94** having a shaft **96** extending therefrom to support a main body **98** having male coupler **70** on one side and a female coupler **72** on the opposing side. Mount **30** suspends module **20** in spaced relation to support pole **12** to which mount **30** is attached. Male coupler **70** and female coupler **72** are configured in same manner as described above with respect to module **20**, i.e., male coupler **70** includes an end face **86** having concentric ring contacts **88** and female coupler **72** has brush contacts **84** positioned within receptacle **82**. Male coupler further includes flange **74** and female coupler **72** includes flange **80**. As a result, module **20** may be coupled to mount **30** in the same manner as described above with respect to the connection of module **20** *a* to module **20** *b*.

[0057] Referring to FIG. **13**, joining of mount **30** to module **20** allows coupler circuitry **110** of female coupler **72** of mount **30** to connect with coupler circuitry **112** of male coupler **70** of module **20** via brush contacts **84** and ring contacts **88**. Coupler circuitry **112** is coupled to LED circuitry **118** via cabling **114**. LED circuitry **118** is also coupled to coupler circuitry **110** associated with female coupler **72** via cabling **114**. As a result, independent power pathways for LED circuitry **118** *b* extend through module **20** from mount **30** and are available at coupler **70** such that a power supply connected to coupler **72** will also provide power to coupler **70**. Similarly, module **20** may also be connected to the male coupler **70** of mount **30** using female coupler **72** of module **20**, thus simply reversing the connections of FIG. **13** such that power is provided by mount **30** to coupler **72** with the power also made available at coupler **70** for attachment of another module **20**.

[0058] Referring to FIG. **14**, cylindrical bearing surfaces of male coupler **70** and female coupler **72** allows adjacent lighting modules **20**, as well as lighting modules **20** coupled to mount **30**, to be rotated about longitudinal axis X-X. The orientation of the rectangular illumination provided by module **20** may thus be adjusted in a single direction, i.e., about a single axis, via rotation of lighting module **20** about axis X-X. As explained above, bearing surfaces **76** and **77** allow for physical rotation of housings **20**, with brush contacts **84** and ring contacts **88** maintaining electrical continuity regardless of the rotation of housing about longitudinal axis X-X. Housings **20** may thus be easily oriented, or reoriented, as desired. While housings **20** may be manually adjusted at any time, servo motors could be incorporated into couplers **70** and **72** to allow for remote rotation of lighting modules **20** about axis X-X.

[0059] Referring to FIGS. **15** and **16**, controller stack **24** comprises a series of core enclosures **132**, each of which houses the power conversion and LED electronics, typically referred to as LED drivers, for an associated lighting module **20**, as well as a master enclosure **140** that provides housekeeping functions. Controller stack **24** includes a back plane **134** that provides the electrical interconnections between each core enclosure **132** and master enclosure **140** as well as the requisite interconnections to wiring harness **22** to interconnect controller stack **24** to lighting modules **20**. Back plane **134** is preferably adapted to act as a heat sink and transfer excess heat to support pole **12** for additional dispersion of heat generated by controller stack **24**. As seen in FIG. **16**, core enclosure **132** and/or master enclosure **140** include ribs **136** for dissipation of heat generated by internal electrical components positioned in a central cavity **138**.

[0060] Referring to FIG. **17**, each core enclosure **132** *a*, **132** *b* . . . **132** *n* is associated with and coupled via wiring harness **22** to a corresponding lighting module **20** *a*, **20** *b* . . . **20** *n*. Preferable, a backup core enclosure **132** *z* is selectively coupled to each lighting module **20** *a*, **20** *b* . . . **20** *n* via a switching circuit **133** to provide a backup power supply in the event of a fault in any of core enclosure **132** *a*, **132** *b* . . . **132** *n*. For example, if a fault in any core enclosure **132** results in the loss of illumination from any or all of the independently controlled rows **50** of LED sets **52** in the corresponding lighting module **20**, power to that lighting module **20** can be switched to the backup core enclosure **132** *z* to maintain the desired amount of illumination until such time as the faulty core enclosure **132** can be repaired or replaced. Each core enclosure **132** *a*, **132** *b* . . . **132** *n* is also interconnected to master enclosure **140**, which supervises and controls via digital commands the local operation of each core enclosure **132** *a*, **132** *b* . . . **132** *n*.

[0061] Referring to FIG. **18**, master enclosure **140** is coupled to AC power via a power and signal connector **158** and includes local AC/DC conversion **142** with input power monitoring **144** as well as surge protection and waveform correction **146**. Master enclosure **140** also includes a controller/processor **148** that has sensor inputs **150** for monitoring of system **10**. Controller/processor **148** is also interconnected to a series of expansion headers **152** and wireless communication interface **156** via a field programmable gate array (FPGA) **154**.

[0062] Controller/processor **148** may thus be programmed to establish connection with a remotely positioned host system or remote device (such as a tablet or smartphone) that can provide commands controlling operation of lighting modules **20** using expansion headers **152** to provide the desired wireless connectivity. Communication could comprise any conventional wireless communication technology or protocol, such as WiFi, Blutetooth®, BLE, ZigBee, Z-Wave, 6lo WPAN, NFC, cellular such as 4G, 5G or LTE, RFID, LoRA, LoRaWAN, Sigfox, NB-IoT, or LIDAR. Controller/processor **148** is also coupled via power and signal connector **158** for communication with core enclosures **132**, such as via a general-purpose input/output (GPIO) line **160**, extending in back plane **134**.

[0063] Referring to FIG. **19**, each core enclosure **132** includes a power and signal connector **170**, which provides connectivity to master enclosure **140** via GPIO line **160** as well as to a connection to AC power. Core enclosure **132** provides power conversion to DC and power conditioning via an EMI filter **172**, an inrush protection circuit **174** and an active power factor corrector (PFC) **176**. A plurality of isolated DC/DC circuits **178**, each of which supports a corresponding one of independently controllable LED rows of asymmetric illumination source **44**, are coupled to active PFC **176**. The present invention is illustrated with three isolated DC/DC circuits because the exemplary illumination source **44** has three independently powered rows of LEDs, but if asymmetric illumination source **44** included four independently controlled rows **50** of LED sets **52**, four isolated DC/DC circuits **178** would be included. Core enclosure **132** further comprises an isolated auxiliary output **180** coupled to a microprocessor **182**. Microprocessor **182** is further coupled to primary sensing circuits **184** and secondary sensing circuits **186** for monitoring voltage, current, power factor, and temperature across system **10**. Microprocessor **182** is further configured to adjust the power output from each of the plurality of isolated DC/DC circuits **178** based on monitoring of primary sensing circuits **184** and secondary sensing circuits **186**. For example, if one of independently controlled rows **50** of LED sets **52** is not operational, microprocessor **182** can adjust the power output from the isolated DC/DC circuits **178** for the other of the independently controlled rows **50** of LED sets **52** to compensate for the loss and ensure that asymmetric illumination source **44** is providing the desired amount of illumination. Referring to FIG. **20**, the wireless communication capability of master enclosure **140** provides a third layer of redundancy in the event of a partial or total loss of illumination from lighting module **20**. For example, a detected loss at one location of system **10** may be communicated to wireless gateway **190** and remote host **192**. The illumination output of another system **10** *b* may then be adjusted accordingly, either by allowing a user to send a command to system **10** *b* to adjust power to lighting modules **20** to compensate for the detected loss or by supervisory software residing on host **192** that automatically sends the appropriate commands.

[0064] Referring to FIG. **21**, asymmetric illumination source **44** of each module **20** allows for remote beam steering of lighting system **10**. Lighting system **10** may be adapted to a particular installation regarding of the width of the pitch to be illuminated, the height of support pole **12**, and the distance between support pole **12** and the targeted pitch. For example, asymmetric illumination source **44** may be driven to change the beam angle (generally recognized as the region of illumination with at least fifty percent of the maximum beam strength) to provide the appropriate amount of illumination between a minimum and maximum spread angle encountered in an installation. In the first scenario of FIG. **19**, where the height of support pole **12** and setback distance require a minimum spread angle, asymmetric illumination source **44** can be driven asymmetrically in a first configuration to provide a narrow beam angle without having to physically reorient modules **20**. In the last scenario, where the height of pole **12** and setback distance require a minimum spread angle, asymmetric illumination source **44** can be driven asymmetrically in a different configuration to provide a broader spread angle without having to physically reorient modules **20**. Thus, the effective positioning of modules **20** can be adjusted without actually having to physically reorient modules **20**. Thus, modules **20** may be asymmetrically driven to change the illumination scenario for different events or conditions, or to simply adjust the illumination in a given location without having to physically move lighting modules **20**. FIG. **20** illustrates how the power control over each row **50** of asymmetric illumination source **44** can be adjusted to impact the beam angle emitted from lighting module **20** without having to rotate lighting module **20**.

[0065] Referring to FIG. **23**, asymmetric illumination source **44** of each lighting module **20** provides for a tunable cut-off for the illumination generated from lighting module **20**.

[0066] Illumination cut-off generally refers to the amount of illumination in the beam field that extends beyond the desired beam angle (any area of illumination with less than fifty percent but more than ten percent of the maximum beam strength). For example, in the first scenario of FIG. **23**, the cut-off is very sharp, i.e., there is very little spillage beyond the main beam angle. In the second and third scenarios, the spillage increases such that more illumination is provided ancillary to the primary beam angle. Asymmetric illumination source **44** may be driven to change the cut-off at any time, whether finally upon installation, or dynamically over time to change the lighting scheme as desired by a user for different applications. For example, a gradual cut-off may be selected when more light is desired in the areas surrounding a pitch for a particular event, such as a pre-game show, and then adjusted to provide a sharp cut-off during a game. Thus, asymmetric illumination source **44** allows for control over both the beam angle and the beam field relative to each other and relative to the illumination target.

[0067] Referring to FIG. **24**, lighting module **20** may be constructed using a housing **240** that encloses an asymmetric illumination source **244** and is environmentally sealed prior to attachment of lens array **260**. As seen in FIG. **25**, housing **240** includes a resilient optical layer **248** positioned over asymmetric illumination source **244** and captured within rectangular opening **242** to seal housing **240** from environmental infiltration. As a result, lens array **260** may be attached or removed from housing **240** in the field, such as to adjust the optical conditioning being provided, without compromising the environmental integrity of housing **240**. Optical layer **248** is preferably formed from a moldable optical silicone, such as SILASTIC® MS-1002 moldable silicone and related moldable silicone compounds. As seen in FIG. **26**, optical layer **248** may include micro-lenses **262** molded therein and in alignment with each LED set **252** of asymmetric illumination source **244**. Optical layer **248** thus performs pre-modulation of the illumination from lighting module **20**. Micro-lenses **262** allow for finer optical texturing than with lens array **260** alone. In addition, as lens array **260** does not need to perform as much optical conditioning, lens array **260** can be smaller and thus lighter than otherwise possible.

[0068] Referring to FIGS. **27** through **30**, lighting module **20** may be outfitted with lens array **60** configured that steers illumination into three, four, or five different regions. For example, each particular installation may include a different number of support poles **12**, so an appropriate lens array **60** distributing illumination into three, four, or five different regions may be used. As is known in the field, illumination from each support pole **12** may need to overlap with illumination for other support poles **12** to provide the desired illumination, reduce or control shadowing, etc. As seen in FIG. **30**, lighting module **20** can provide a wide or narrow area of illumination using variously designed lens arrays **60** to steer illumination between a minimum and maximum distribution angle.

[0069] System **10**, having lighting modules **20** and emergency auxiliary lighting **200** may lower demand for electricity by up to 80% over existing lighting systems. Battery **210** may incur a draw on system **10** but once battery **210** is fully charged, an inactive emergency system required negligible power. This lower demand may exist even with the current grid infrastructure to the stadium where system **10** is installed. By installing electrical vehicle charging systems in the stadium parking lot, electrical demand may be monetized by the stadium owner during low-demand periods or during stadium use.

[0070] The electrical vehicle charging system ("EV System") may include a plurality of electrical vehicle chargers ("EV Chargers") installed in a stadium parking lot and connected to existing infrastructure. Due to decreased power usage for stadium lighting system **10**, the current electrical grid may be used without the need to install transformers or power control equipment.

[0071] In certain embodiments, lighting system **10** may be placed in the stadium parking lot, to provide lighting in the parking area, and modular EV Chargers may be connected to pole **12**. In certain other embodiments, the EV Chargers may be connected to existing lighting poles in the parking lot or left as free standing devices for parking spaces not near lighting posts. Such charging stations may have charging cables with connectors compatible with standard electrical vehicle charging ports in the marketplace.

[0072] The EV Chargers may have two-way remote communication made possible through internet of things modules (IoT modules) offering positioning and messaging communication between the lighting System **10** and the EV System, including individual EV Chargers. IoT modules may include communication chips or services such as UBlox IoT modules or similar products and services.

[0073] System **10** may use the wireless gateway **190** to communicate with the EV System. The master controller **140** for System **10** may also act as the controller for the EV System.

[0074] In certain other embodiments a system controller may be present to coordinate power distribution between System **10** and the EV System through a smart gateway. The wireless gateway **190** may then continue to coordinate wireless communication within System **10** and the smart gateway may be used to coordinate power distribution and communication between System **10** and the EV System.

[0075] The IoT Modules within the EV Chargers may be used to send and receive wireless signals to manage and control power usage and to monitor the health and operation of each of the EV Chargers. The data exchanged by the signals may also lower or increase power output if the System **10** requires more power from the electrical grid.

[0076] A user interface on each of the EV Chargers may provide for local device control of each of the EV Chargers. The user interface may also allow customers to interact with an individual EV Charger. The Smart Gateway **190** may also allow users via an app on their phone or mobile computing device to communicate and interact with the EV Charger. For example, the EV Charger may have two-way communication with the electrical vehicle being charged and may provide information on the charging status of the vehicle. Since parking at sporting events may involve thousands of vehicles, the charging station may be able to provide location data for customers to find their vehicles after the event.

[0077] The EV Chargers may also have a payment system integrated with the user interface or accessible by the user interface. The user interface may also have alarms to signal the user if charging rises above a threshold value or if there is some kind of error or fault within the device.

[0078] As part of a health-status check, the system controller may receive error or fault communications from the EV Chargers. A management user interface on a computing device, phone, or mobile computing device may be used to receive and display, location and health-information of the EV Chargers. The management user interface may also be configured to provide control of the EV Chargers and reset and repair functions.

[0079] Other two-way controllable features may also be present to provide for system control and feedback.

[0080] The modular configuration of the EV Charger may provide for addition or subtraction of features and components of the EV Charger, forming a configurable EV Charging System. The Charging System may be configured or expanded based on, for example, the location of a stadium or venue, the current grid infrastructure, or even the event at the stadium of the venue.

[0081] The EV Charging System may have a storage battery or storage batteries providing for charging in, for example, emergency situations or situations where draw is at the capacity of the grid infrastructure. In the event of an emergency, it may also be possible to configure the EV Chargers to draw power from storage battery and/or the electrical vehicle being charged, to provide lighting to the parking lights or the stadium lights. An emergency may include a black-out or a brown-out. The emergency may also be signaled by the lighting system where the controller stack provides a signal from the wireless gateway **190** to the internet of things module

[0082] Each of the EV Chargers may have its own battery. The modular configuration may provide for configurations where parts from one EV Charger are used to repair another charger.

[0083] In a certain embodiment, a lighting system, having at least one lighting module, the lighting module having a housing extending from first end to a second end along a longitudinal axis and having an elongated opening in alignment with an illumination source within. A first coupler may be positioned at the first end of the housing and may have an end face having a first set of electrical contacts. The first coupler may also have a first set of auxiliary electrical contacts. A second coupler may be positioned at the second end of the housing and have an internal bore. The internal bore may have a second set of electrical contacts and a second set of auxiliary electrical contacts within. An auxiliary lighting module may have a housing having an opening and an illumination source within. A coupler connected to the auxiliary lighting module housing may have a set of auxiliary module electrical contacts and the coupler may be configured for connection to one of the first coupler or the second coupler. A local power source may be connected to a controller stack. The controller stack may be connected to the at least one lighting module by a first set of wiring, and connected to the auxiliary lighting module by a second set of wiring. The controller stack may have an alternating current power supply, a wireless, and a switching circuit. The switching circuit may have a voltage monitor connecting the alternating current power supply and the first set of wiring, and the local power source and the second set of wiring. The voltage monitor may measure for a voltage drop based on a threshold value. The first set of electrical contacts may be connected to the second set of electrical contacts internally, within the lighting module housing. The second set of wiring may connect the first set of auxiliary electrical contacts and the second set of auxiliary electrical contacts. The lighting system may be connected to the pole **12**. A plurality of lighting systems may be connected to a plurality of poles. A plurality of electrical vehicle chargers may be connected to the controller stack of each of the plurality of lighting systems. Each of the plurality of electrical vehicle chargers may have: an electrical vehicle charging cable configured for connection to an electrical vehicle, a connection to an existing power grid, a modular configuration, an internet of things module configured for two-way, wireless communication with a wireless gateway **190**, a feedback system, and a control system connected to and configured for communication with the internet of things module, the existing power grid, and the electrical vehicle charging cable. The electrical charger may have a user interface configured to receive commands to control the electrical charger and to provide display feedback to a user.

[0084] In certain embodiments, the power supply may be a direct current power supply. In certain other embodiments, the power supply may be an alternating current power supply, and the controller stack may have a rectifier to provide direct current (DC) to any drivers or circuitry within the controller stack and to provide DC to the lighting modules.

[0085] As described above, the present invention may be a system, a device, a method, and/or a computer program associated therewith and is described herein with reference to flowcharts and block diagrams of methods and systems. The flowchart and block diagrams illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer programs of the present invention. It should be understood that each block of the flowcharts and block diagrams can be implemented by computer readable program instructions in software, firmware, or dedicated analog or digital circuits. These computer readable program instructions may be implemented on the processor of a general purpose computer, a special purpose computer, or other programmable data processing apparatus to produce a machine that implements a part or all of any of the blocks in the flowcharts and block diagrams. Each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical functions. It should also be noted that each block of the block diagrams and flowchart illustrations, or combinations of blocks in the block diagrams and flowcharts, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

What is claimed is:

1. An electrical vehicle charging system comprising:

a lighting system comprising:

a controller stack connected to:

a lighting module and an auxiliary lighting module by wiring; and

a wireless communication gateway

an alternating current power supply;

a switching circuit;

and a local power storage device;

a plurality of electrical vehicle chargers each comprising:

an electrical vehicle charging cable configured for connection to an electrical vehicle;

a connection to an existing power grid;

an internet of things module configured for two-way, wireless communication with a wireless gateway;

a feedback system;

a control system connected to and configured for communication with the internet of things module, the existing power grid, and the electrical vehicle charging cable.

2. The lighting system of claim **1**, wherein each electrical vehicle charger has a modular configuration.

3. The lighting system of claim **1**, wherein the local power storage device is a battery.

4. The lighting system of claim **1**, wherein each of the plurality of electrical vehicle chargers have a local power storage device, and the local power storage device is connected to the existing power grid.

5. The lighting system of claim **1**, wherein the local power storage device of each of the plurality of electrical vehicle chargers is a battery.

6. The lighting system of claim **1**, wherein a user interface configured to receive input and send commands to control the electrical charger and to provide display feedback to a user.

7. The lighting system of claim **1**, wherein the electrical vehicle charging cable is configured to provide power to the electrical vehicle when connected to the electrical vehicle.

8. The lighting system of claim **1**, wherein the electrical vehicle charging cable is configured to draw power from the electrical vehicle to the existing power grid when connected to the electrical vehicle and an emergency signal is received by the internet of things module.

9. The lighting system of claim **4**, wherein local power storage device provides power to the existing power grid when an emergency signal is received by the internet of things module.

10. The lighting system of claim **9**, wherein the electrical vehicle charging cable is configured to draw power from the electrical vehicle to the existing power grid when connected to the electrical vehicle and the emergency signal is received by the internet of things module and the local power storage device power output falls below a defined threshold.

11. The lighting system of claim **1**, wherein internet of things module sends and receives signals to manage and control power usage.

12. The lighting system of claim **1**, wherein the internet of things module sends and receives signals to monitor the operation of each of the plurality of electrical vehicle chargers.

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